

# Economics II

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# 1 The Foundations of Index Numbers

## 1.1 What is an Index Number?

An Index Number (IN) is a statistical device used to compare the general magnitude of a group of distinct, but related, variables across different situations (typically different time periods). We always compare a **current/target period** against a **base/reference period**.

### Definition 1: Price Relative

A price relative is the ratio of the price of a single commodity in the current period ( $p_1$ ) to its price in the base period ( $p_0$ ). It is often expressed as a percentage:

$$P_{01} = \frac{p_1}{p_0} \times 100 \quad (1)$$

### 1.1.1 Properties of Price Relatives

For periods  $i$ ,  $j$ , and  $k$ , price relatives obey three strict mathematical properties:

- **Identity Property:**  $P_{ii} = 1$
- **Time Reversal Property:**  $P_{ij} \times P_{ji} = 1$
- **Cyclical (Circular) Property:**  $P_{ij} \times P_{jk} \times P_{ki} = 1$

## 1.2 Value and Quantity Relatives

Just as we measure prices, we measure volumes (quantities) and total values.

- **Quantity Relative:**  $Q_{01} = \frac{q_1}{q_0}$
- **Value Relative:** Since total value is price  $\times$  quantity ( $V = pq$ ), the value relative is the product of the price and quantity relatives:  $V_{01} = P_{01} \times Q_{01}$ .

# 2 Construction and Major Formulae

## 2.1 The 7 Steps of Construction

Before calculating an aggregate index, a statistician must systematically define the problem:

1. **Define the Purpose:** e.g., Wholesale Price Index (WPI) vs. Consumer Price Index (CPI).
2. **Choice of Base Period:** Must be a "normal" economic period, not too distant in the past.
3. **Collection of Data:** Gathering accurate price quotations.
4. **Choice of Commodities:** Using judgment sampling to select a representative basket.
5. **Method of Combining Data:** Choosing arithmetic means, geometric means, or aggregates.
6. **Choice of Weights/Formula:** Assigning economic importance (usually quantities) to prices.
7. **Interpretation:** Translating the index into real-world purchasing power changes.

## 2.2 Weighted Aggregative Indices

To avoid the flaws of a simple unweighted index (which skews based on units of measurement), we apply quantity weights to our prices. Let  $p_0, q_0$  be base period prices and quantities, and  $p_1, q_1$  be current period prices and quantities.

### Definition 2: Laspeyres' Index

Uses **base period quantities** ( $q_0$ ) as fixed weights. It measures the current cost of the exact basket of goods purchased in the base year.

$$L_{01} = \frac{\sum_{i=1}^n p_1^i q_0^i}{\sum_{i=1}^n p_0^i q_0^i} \quad (2)$$

### Definition 3: Paasche's Index

Uses **current period quantities** ( $q_1$ ) as weights. It measures how much today's exact consumption habits would have cost in the base year.

$$P_{01} = \frac{\sum_{i=1}^n p_1^i q_1^i}{\sum_{i=1}^n p_0^i q_1^i} \quad (3)$$

### Definition 4: Fisher's Ideal Index

The geometric mean of Laspeyres' and Paasche's indices. It mathematically balances the upward bias of Laspeyres and the downward bias of Paasche.

$$F_{01} = \sqrt{L_{01} \times P_{01}} = \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0} \times \frac{\sum p_1 q_1}{\sum p_0 q_1}} \quad (4)$$

## 3 Fisher's Tests of Adequacy

Irving Fisher proposed mathematical tests to determine if an index number formula is logically sound.

### Theorem 1: Time Reversal Test (TRT)

An index number should work symmetrically forward and backward in time. The product of the forward index and the backward index must equal unity.

$$I_{01} \times I_{10} = 1 \quad (5)$$

*Application:* Laspeyres and Paasche fail this test. Fisher's Ideal Index passes.

### Theorem 2: Factor Reversal Test (FRT)

The product of a Price Index and a Quantity Index (calculated using the exact same formula structure) must equal the true Value Index ( $V_{01} = \frac{\sum p_1 q_1}{\sum p_0 q_0}$ ).

$$P_{01} \times Q_{01} = V_{01} \quad (6)$$

*Application:* Laspeyres and Paasche fail this test. Fisher's Ideal Index passes.

### Theorem 3: Circular Test

A generalized, multi-period version of the Time Reversal Test. If indices are chained across multiple periods and loop back to the base period, their product must equal unity.

$$I_{01} \times I_{12} \times I_{23} \cdots \times I_{n0} = 1 \quad (7)$$

*Application:* Fisher's Ideal Index **fails** this test due to its shifting current-year weights.

## 4 Practice Problem: Calculating Index Numbers

**Problem Statement:** Calculate the Laspeyres, Paasche, and Fisher Ideal Price Index numbers for Year 1 (using Year 0 as the base) from the following data. Express your final answers as percentages.

| Commodity | Base Price ( $p_0$ ) | Base Quantity ( $q_0$ ) | Current Price ( $p_1$ ) | Current Quantity ( $q_1$ ) |
|-----------|----------------------|-------------------------|-------------------------|----------------------------|
| A         | 10                   | 5                       | 15                      | 4                          |
| B         | 20                   | 2                       | 25                      | 3                          |
| C         | 5                    | 10                      | 8                       | 8                          |

### Solution Step 1: The Calculation Table

Before applying any formulae, we construct a table to find the four required aggregate sums:  $\sum p_0 q_0$ ,  $\sum p_1 q_0$ ,  $\sum p_0 q_1$ , and  $\sum p_1 q_1$ .

| Commodity                        | $p_0 q_0$          | $p_1 q_0$          | $p_0 q_1$          | $p_1 q_1$          |
|----------------------------------|--------------------|--------------------|--------------------|--------------------|
| A                                | $10 \times 5 = 50$ | $15 \times 5 = 75$ | $10 \times 4 = 40$ | $15 \times 4 = 60$ |
| B                                | $20 \times 2 = 40$ | $25 \times 2 = 50$ | $20 \times 3 = 60$ | $25 \times 3 = 75$ |
| C                                | $5 \times 10 = 50$ | $8 \times 10 = 80$ | $5 \times 8 = 40$  | $8 \times 8 = 64$  |
| <b>SUM (<math>\Sigma</math>)</b> | <b>140</b>         | <b>205</b>         | <b>140</b>         | <b>199</b>         |

### Solution Step 2: Laspeyres' Price Index ( $L_{01}$ )

Using base-year quantities as weights:

$$\begin{aligned} L_{01} &= \frac{\sum p_1 q_0}{\sum p_0 q_0} \times 100 \\ L_{01} &= \frac{205}{140} \times 100 \\ L_{01} &= 1.4643 \times 100 = \mathbf{146.43} \end{aligned}$$

### Solution Step 3: Paasche's Price Index ( $P_{01}$ )

Using current-year quantities as weights:

$$\begin{aligned} P_{01} &= \frac{\sum p_1 q_1}{\sum p_0 q_1} \times 100 \\ P_{01} &= \frac{199}{140} \times 100 \\ P_{01} &= 1.4214 \times 100 = \mathbf{142.14} \end{aligned}$$

### Solution Step 4: Fisher's Ideal Price Index ( $F_{01}$ )

Taking the geometric mean of Laspeyres and Paasche:

$$\begin{aligned}F_{01} &= \sqrt{L_{01} \times P_{01}} \\F_{01} &= \sqrt{1.4643 \times 1.4214} \times 100 \\F_{01} &= \sqrt{2.0813} \times 100 \\F_{01} &= 1.4427 \times 100 = \mathbf{144.27}\end{aligned}$$

Note that Fisher's index (144.27) logically falls strictly between Laspeyres (146.43) and Paasche (142.14).

## 5 Chain Indices and Statistical Errors

### 5.1 The Chain Index

Fixed-base indices become obsolete over time as consumer tastes change and new commodities enter the market. To solve this, statisticians use a **Chain Index**, which constantly updates the base to the immediately preceding period.

#### Definition 5: Chain Index

A long-term index constructed by multiplying a series of short-term *link indices* together.

$$I_{0n} = I_{01} \times I_{12} \times I_{23} \times \cdots \times I_{n-1,n} \quad (8)$$

*Note:* To shift a base period from  $t_0$  to  $t_1$  (a process called **Splicing**), the formula is:

$$I_{t_1 t} = \frac{I_{t_0 t}}{I_{t_0 t_1}} \times 100.$$

### 5.2 Errors in Index Numbers

Index numbers are subject to three primary types of statistical errors:

- **Formula Error:** Arises from the choice of formula. Estimated by the difference between Laspeyres and Paasche ( $D = L_{01} - P_{01}$ ).
- **Sampling Error:** Arises because we use a sample ( $n$ ) to represent the population ( $N$ ). Reduced by using *stratified sampling* instead of simple random sampling.
- **Homogeneity Error:** Arises from the appearance of *unique commodities* (items existing in only one period) and the disappearance of old ones. Measured by the ratio  $R$ :

$$R = \frac{N_1 + N_0}{N_1 + N_0 + N_{01}} \quad (9)$$

Where  $N_1$  and  $N_0$  are unique commodities, and  $N_{01}$  represents binary (shared) commodities.

## 6 Real-World Applications

### 6.1 Deflation and Real Income

Inflation decreases the purchasing power of money. To find the true economic value of current income relative to a base year, we **deflate** it using the Consumer Price Index (CPI).

$$\text{Real Income} = \frac{\text{Current Income}}{\text{CPI}} \quad (10)$$

## 6.2 Wholesale Price Index (WPI) vs. Consumer Price Index (CPI)

### WPI vs. CPI Comparison

#### Wholesale Price Index (WPI):

- Measures inflation of bulk goods in primary/wholesale markets.
- Calculation is a two-stage process: Uses **Jevons Index (Geometric Mean)** for elementary items, and **Weighted Arithmetic Mean** for higher groups.

#### Consumer Price Index (CPI):

- Measures inflation of retail goods/services bought by households.
- Uses the **Laspeyres Formula** (fixed base-year quantities).
- Used to regulate Dearness Allowances (DA) and approximate the Cost of Living.

## 6.3 The True Cost of Living Index (CLI)

The True CLI measures the cost of maintaining an exact level of *utility* (satisfaction). Because consumers substitute cheaper goods when prices rise, the fixed-basket **Laspeyres index overestimates** the true CLI, while the **Paasche index underestimates** it.

## 7 Index of Industrial Production (IIP)

### 7.1 Quantum of Production and Value Added

The IIP measures the short-term changes in the physical volume (quantum) of industrial production (Mining, Manufacturing, Electricity).

To avoid **double counting** (e.g., counting both the steel and the car made from it), we must measure **Value Added**.

$$\text{Value Added} = \text{Total Output Value} - \text{Total Input Value}$$

### 7.2 Mathematical Formulation

Let capital letters ( $P, Q$ ) represent **Output** and lowercase letters ( $p, q$ ) represent **Input**. The ratio of Value Added ( $V_{01}$ ) is:

$$V_{01} = \frac{\sum P_1 Q_1 - \sum p_1 q_1}{\sum P_0 Q_0 - \sum p_0 q_0} \quad (11)$$

The **Laspeyres Form** ( $I_L$ ) for the Index of Industrial Production uses base-year prices as weights:

$$I_L = \frac{\sum P_0 Q_1 - \sum p_0 q_1}{\sum P_0 Q_0 - \sum p_0 q_0} \quad (12)$$

### 7.3 Methods of Computation

Because calculating the exact input/output formula for an entire economy is too complex, statisticians use two proxy methods:

1. **Method of Deflation:** Divides current values by a Price Index to estimate base-year costs.

2. **Method of Indicators:** The most practical method. Calculates a simple production indicator ( $I_{01}$ ) for small sub-sectors, then aggregates them using base-year Value Added ( $V_0$ ) as the weight:

$$N_{01} = \frac{\sum I_{01} V_0}{\sum V_0} \quad (13)$$

## 8 Practice Problem: Index of Industrial Production (IIP)

**Problem Statement:** A manufacturing firm produces two output goods (Capital letters) and uses two raw material inputs (Lowercase letters). Calculate the Index of Industrial Production for Year 1 (using Year 0 as the base) using the Laspeyres form. Express your final answer as a percentage.

| Item                | Type              | Base Price<br>( $P_0$ or $p_0$ ) | Base Qty<br>( $Q_0$ or $q_0$ ) | Current Qty<br>( $Q_1$ or $q_1$ ) |
|---------------------|-------------------|----------------------------------|--------------------------------|-----------------------------------|
| Output A (Tractors) | Output ( $P, Q$ ) | 200                              | 5                              | 6                                 |
| Output B (Cars)     | Output ( $P, Q$ ) | 100                              | 10                             | 15                                |
| Input X (Steel)     | Input ( $p, q$ )  | 20                               | 30                             | 40                                |
| Input Y (Rubber)    | Input ( $p, q$ )  | 10                               | 50                             | 70                                |

### Solution Step 1: State the Formula

The Laspeyres form for the Index of Industrial Production (IIP) uses base-year prices as weights to calculate the ratio of Value Added:

$$I_L = \frac{\sum P_0 Q_1 - \sum p_0 q_1}{\sum P_0 Q_0 - \sum p_0 q_0} \times 100 \quad (14)$$

### Solution Step 2: The Calculation Table

We must calculate the total value of outputs and inputs for both periods, strictly holding prices constant at the base year ( $P_0$  and  $p_0$ ).

| Item                                      | Base Year Values ( $P_0 Q_0$ or $p_0 q_0$ ) | Current Year at Base Prices ( $P_0 Q_1$ or $p_0 q_1$ ) |
|-------------------------------------------|---------------------------------------------|--------------------------------------------------------|
| <b>Outputs (<math>P, Q</math>)</b>        |                                             |                                                        |
| Output A                                  | $200 \times 5 = 1000$                       | $200 \times 6 = 1200$                                  |
| Output B                                  | $100 \times 10 = 1000$                      | $100 \times 15 = 1500$                                 |
| <b>Total Output (<math>\Sigma</math>)</b> | <b>2000</b>                                 | <b>2700</b>                                            |
| <b>Inputs (<math>p, q</math>)</b>         |                                             |                                                        |
| Input X                                   | $20 \times 30 = 600$                        | $20 \times 40 = 800$                                   |
| Input Y                                   | $10 \times 50 = 500$                        | $10 \times 70 = 700$                                   |
| <b>Total Input (<math>\Sigma</math>)</b>  | <b>1100</b>                                 | <b>1500</b>                                            |

### Solution Step 3: Compute Value Added

Now we subtract the total inputs from the total outputs for both the denominator (Base Year) and numerator (Current Year).

$$\begin{aligned} \text{Base Year Value Added (Denominator)} &= \sum P_0 Q_0 - \sum p_0 q_0 \\ &= 2000 - 1100 = \mathbf{900} \end{aligned}$$

$$\begin{aligned}\text{Current Year Value Added (Numerator)} &= \sum P_0 Q_1 - \sum p_0 q_1 \\ &= 2700 - 1500 = \mathbf{1200}\end{aligned}$$

#### Solution Step 4: Final Index Calculation

Plug the Value Added totals back into the main formula:

$$\begin{aligned}I_L &= \frac{1200}{900} \times 100 \\ I_L &= 1.3333 \times 100 = \mathbf{133.33}\end{aligned}$$

*Interpretation: The physical volume of industrial production (Value Added) for this firm has increased by 33.33% in Year 1 compared to Year 0.*

## 9 The Axiomatic Approach to Price Indices

In pure mathematical statistics, a price index is viewed as a positive-valued function mapping the price and quantity vectors of a base and current year into a single scalar:  $P : \mathbb{R}_{++}^{4n} \rightarrow \mathbb{R}_{++}$ .

### 9.1 The Five Fundamental Axioms

For  $P(p_0, q_0, p_1, q_1)$  to be a valid index, it must satisfy:

1. **Monotonicity:** Strictly increasing w.r.t  $p_1$  and strictly decreasing w.r.t  $p_0$ .
2. **Linear Homogeneity:**  $P(p_0, q_0, \lambda p_1, q_1) = \lambda P(p_0, q_0, p_1, q_1)$ .
3. **Identity:**  $P(p_0, q_0, p_0, q_1) = 1$ .
4. **Dimensionality:**  $P(\lambda p_0, q_0, \lambda p_1, q_1) = P(p_0, q_0, p_1, q_1)$ .
5. **Commensurability:** Invariant to changes in physical units of measurement.

### 9.2 Proofs of Axiom Satisfaction

#### Theorem 4: Laspeyres Satisfies All Axioms

Let  $L(p_0, q_0, p_1, q_1) = \frac{\sum p_1 q_0}{\sum p_0 q_0}$ .

- **Monotonicity:** If  $p_1 > \bar{p}_1$ , then  $\sum p_1 q_0 > \sum \bar{p}_1 q_0$ , thus  $L$  increases.
- **Homogeneity:**  $L(p_0, q_0, \lambda p_1, q_1) = \frac{\sum (\lambda p_1) q_0}{\sum p_0 q_0} = \lambda \frac{\sum p_1 q_0}{\sum p_0 q_0} = \lambda L$ .
- **Identity:**  $L(p_0, q_0, p_0, q_1) = \frac{\sum p_0 q_0}{\sum p_0 q_0} = 1$ .
- **Dimensionality:**  $L(\lambda p_0, q_0, \lambda p_1, q_1) = \frac{\sum (\lambda p_1) q_0}{\sum (\lambda p_0) q_0} = \frac{\lambda \sum p_1 q_0}{\lambda \sum p_0 q_0} = L$ .
- **Commensurability:** Adjusting units scales  $p$  by  $\lambda_i$  and  $q$  by  $1/\lambda_i$ . In the products  $p_1 q_0$  and  $p_0 q_0$ , the scalars cancel out ( $\lambda_i \times 1/\lambda_i = 1$ ), leaving  $L$  unchanged.

### Theorem 5: Paasche Satisfies All Axioms

Let  $P(p_0, q_0, p_1, q_1) = \frac{\sum p_1 q_1}{\sum p_0 q_1}$ .

The proof is identical to Laspeyres, simply substituting  $q_0$  with  $q_1$  in the weights. For example, Identity:  $\frac{\sum p_0 q_1}{\sum p_0 q_1} = 1$ . Homogeneity:  $\frac{\sum (\lambda p_1) q_1}{\sum p_0 q_1} = \lambda P$ .

### Theorem 6: Fisher Satisfies All Axioms

Let  $F = \sqrt{L \times P}$ . Because Fisher is the geometric mean of two indices that perfectly satisfy all five axioms, it inherits these properties mathematically.

*Proof of Homogeneity for Fisher:*

$$\begin{aligned} F(p_0, q_0, \lambda p_1, q_1) &= \sqrt{L(\dots, \lambda p_1, \dots) \times P(\dots, \lambda p_1, \dots)} \\ &= \sqrt{(\lambda L) \times (\lambda P)} = \sqrt{\lambda^2 (L \times P)} = \lambda \sqrt{L \times P} = \lambda F. \end{aligned}$$

## 9.3 Homework Solutions

### Task 1: Prove the Proportionality Test

*Statement:* Every price index satisfying axioms 1-5 satisfies  $P(p_0, q_0, \lambda p_0, q_0) = \lambda$ .

*Proof:* By the Identity Axiom (3), we know  $P(p_0, q_0, p_0, q_0) = 1$ .

By the Homogeneity Axiom (2), we can factor out  $\lambda$  from the current prices:

$$P(p_0, q_0, \lambda p_0, q_0) = \lambda \times P(p_0, q_0, p_0, q_0).$$

Substituting the identity gives:  $\lambda \times 1 = \lambda$ . (Q.E.D)

### Task 2: Examining Proposed Indices A-E

- **(A)**  $I_{01} = \frac{p_1^1}{p_0^1}$ : **Fails** strict Monotonicity. It only accounts for the first commodity; an increase in  $p_1^2$  does not increase the index.
- **(B)**  $I_{01} = \sqrt{\frac{\sum p_1 q_0}{\sum p_0 q_0}}$ : **Fails** Linear Homogeneity. Factoring  $\lambda$  out of the numerator results in  $\sqrt{\lambda} \times I_{01}$ , not  $\lambda \times I_{01}$ .
- **(C)**  $I_{01} = \frac{\sum p_1 q_1}{\sum p_0 q_0}$ : **Fails** Identity. If  $p_1 = p_0$ , the index reduces to a ratio of quantities  $(\frac{\sum q_1}{\sum q_0})$ , which does not equal 1.

## 9.4 Example 1: Calculating Indices for 6 Commodities

Given the following aggregated sums for Rice, Wheat, Jowar, Barley, Bajra, and Gram:

$$\begin{aligned} \sum p_0 q_0 &= 35467, & \sum p_1 q_0 &= 42550 \\ \sum p_0 q_1 &= 49871, & \sum p_1 q_1 &= 60444 \\ \sum p_0 q' &= 42669, & \sum p_1 q' &= 51497 \end{aligned} \quad (\text{where } q' \text{ is the average quantity})$$

$$\text{Laspeyres' Index } (L_{01}): \frac{42550}{35467} \times 100 = \mathbf{119.97}$$

$$\text{Paasche's Index } (P_{01}): \frac{60444}{49871} \times 100 = \mathbf{121.20}$$

$$\text{Edgeworth-Marshall } (EM_{01}): \frac{51497}{42669} \times 100 = \mathbf{120.69}$$

$$\text{Fisher's Ideal Index } (F_{01}): \sqrt{119.97 \times 121.20} = \mathbf{120.58}$$

## 9.5 Example 2: Combining Group Indices

When commodities are categorized into broad economic groups, the overall index is the weighted average of the group indices.

| Group                              | Index ( $I$ ) | Weight ( $W$ %) | Product ( $I \times W$ ) |
|------------------------------------|---------------|-----------------|--------------------------|
| Primary Articles                   | 163.6         | 32.30           | 5284.28                  |
| Fuel, Power etc.                   | 156.6         | 10.66           | 1669.356                 |
| Manufactured Products              | 168.6         | 57.04           | 9616.944                 |
| <b>Total (<math>\Sigma</math>)</b> |               | <b>100.00</b>   | <b>16570.58</b>          |

$$\text{Overall Index } (I_{01}) = \frac{\sum I \times W}{\sum W} = \frac{16570.58}{100} = \mathbf{165.71} \quad (15)$$

## 10 Income and Allied Size Distributions

### 10.1 Size Distributions in Economics

A **Size Distribution** analyzes how an economic variable (like income, wealth, or firm size) is distributed across a population. These distributions are typically strictly non-negative and heavily skewed to the right (e.g., Pareto or Lognormal distributions), meaning a large portion of the population clusters at the lower end while a few outliers exist at the extreme high end.

For a variable to be meaningfully analyzed as a size distribution, it must possess the **Transfer Property**: It must be logically possible to transfer a quantity of the variable from one individual to another (e.g., transferring income is possible; transferring human height is not).

### 10.2 The Lorenz Curve

The Lorenz Curve is the premier graphical method for visualizing economic inequality.

Let  $x$  be income, with probability density function  $f(x)$  and mean  $\mu$ .

- **Cumulative Population Share ( $F$ ):** The proportion of earners with income  $\leq x$ .

$$F(x) = \int_0^x f(t)dt \quad (16)$$

- **Cumulative Income Share ( $F_1$ ):** The proportion of aggregate income held by those earners.

$$F_1(x) = \frac{\int_0^x tf(t)dt}{\int_0^\infty tf(t)dt} = \frac{1}{\mu} \int_0^x tf(t)dt \quad (17)$$

#### 10.2.1 Properties of the Lorenz Curve

1. **Boundaries:** The curve always starts at  $(0,0)$  and ends at  $(1,1)$ .
2. **Monotonicity:** Since  $\frac{dF_1}{dF} = \frac{x}{\mu} > 0$ , the curve is strictly increasing.
3. **Convexity:** Since  $\frac{d^2F_1}{dF^2} = \frac{1}{\mu f(x)} > 0$ , the curve is convex from below. It always sags below the egalitarian Line of Equality.

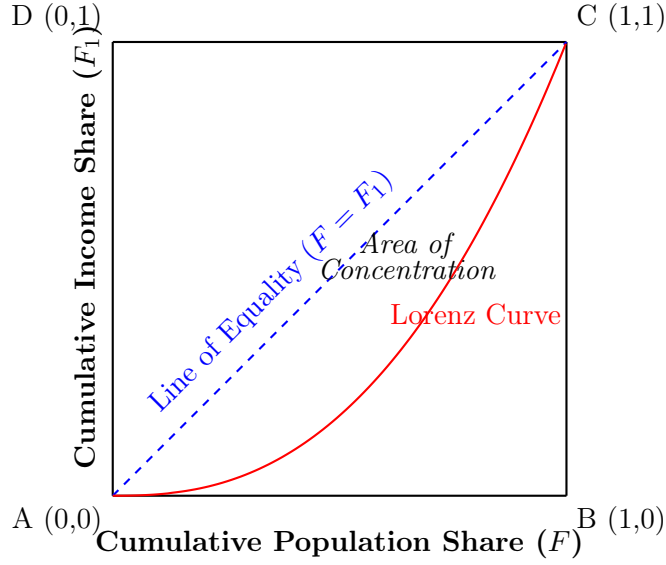


Figure 1: The Lorenz Curve and the Area of Concentration

### 10.3 Gini's Concentration Coefficient (Lorenz Ratio)

To quantify inequality as a single metric, we use the Lorenz Ratio (LR), also known as the Gini Coefficient. It is defined as exactly twice the Area of Concentration.

#### Definition 6: The Lorenz Ratio (Continuous)

The ratio of the Area of Concentration to the total area of the lower triangle ( $\frac{1}{2}$ ).

$$LR = 1 - 2 \int_0^1 F_1 dF \quad (18)$$

*Limits:*  $LR = 0$  indicates perfect equality.  $LR = 1$  indicates perfect inequality (one person holds all wealth).

#### Theorem 7: Numerical Computation (Grouped Data)

For empirical grouped data, where  $p_j$  is the population share of class  $j$ , and  $Q_j$  is the cumulative income share up to class  $j$ , the Lorenz Ratio is approximated by calculating the areas of trapezia under the curve:

$$LR = 1 - \sum_{j=1}^K p_j (Q_j + Q_{j-1}) \quad (19)$$

#### Theorem 8: Numerical Computation (Ungrouped/Discrete Data)

For  $n$  individuals with sorted discrete incomes  $x_1 \leq x_2 \leq \dots \leq x_n$ , the ratio is exactly:

$$LR_1 = \frac{\sum_{r=1}^n (2r-1)x_r}{n^2 \bar{x}} - 1 \quad (20)$$

## 11 Practice Problem: Income Inequality and the Lorenz Ratio

**Problem Statement:** Consider a micro-economy of 5 individuals with the following sorted incomes: Rs. 10, Rs. 20, Rs. 30, Rs. 40, and Rs. 100. Calculate the Lorenz Ratio (Gini Coefficient) using the numerical trapezium method.

### Solution Step 1: Cumulative Data Table

Total Income = Rs. 200. Since  $n = 5$ , each individual represents  $1/5 = 0.20$  of the population ( $p_j$ ).

| Person ( $j$ ) | Income ( $x_j$ ) | Pop. Share ( $p_j$ ) | Cum. Pop. ( $P_j$ ) | Income Share ( $q_j$ ) | Cum. Income ( $Q_j$ ) |
|----------------|------------------|----------------------|---------------------|------------------------|-----------------------|
| 0              | 0                | 0.00                 | <b>0.00</b>         | 0.00                   | <b>0.00</b>           |
| 1              | 10               | 0.20                 | <b>0.20</b>         | $10/200 = 0.05$        | <b>0.05</b>           |
| 2              | 20               | 0.20                 | <b>0.40</b>         | $20/200 = 0.10$        | <b>0.15</b>           |
| 3              | 30               | 0.20                 | <b>0.60</b>         | $30/200 = 0.15$        | <b>0.30</b>           |
| 4              | 40               | 0.20                 | <b>0.80</b>         | $40/200 = 0.20$        | <b>0.50</b>           |
| 5              | 100              | 0.20                 | <b>1.00</b>         | $100/200 = 0.50$       | <b>1.00</b>           |

*Note: The coordinates  $(P_j, Q_j)$  represent the points plotted to draw the Lorenz Curve. For example,  $(0.80, 0.50)$  indicates the poorest 80% of the population holds 50% of the total wealth.*

### Solution Step 2: The Lorenz Ratio ( $LR$ )

We calculate the area under the Lorenz curve using the sum of the areas of the trapezia, and subtract it from 1.

$$LR = 1 - \sum_{j=1}^n p_j(Q_j + Q_{j-1}) \quad (21)$$

Calculating the trapezium area for each interval:

$$j = 1 : 0.20 \times (0.05 + 0.00) = 0.01$$

$$j = 2 : 0.20 \times (0.15 + 0.05) = 0.04$$

$$j = 3 : 0.20 \times (0.30 + 0.15) = 0.09$$

$$j = 4 : 0.20 \times (0.50 + 0.30) = 0.16$$

$$j = 5 : 0.20 \times (1.00 + 0.50) = 0.30$$

Summing these areas gives:  $\sum = 0.01 + 0.04 + 0.09 + 0.16 + 0.30 = \mathbf{0.60}$

$$LR = 1 - 0.60$$

$$LR = \mathbf{0.40}$$

*Conclusion: The Gini Concentration Coefficient (Lorenz Ratio) for this population is 0.40.*

## 12 The Pareto Distribution: Theory and Applications

### 12.1 Empirical Foundations and Pareto's Law

In 1897, the Italian economist Vilfredo Pareto observed a universal regularity in the distribution of income across both capitalist and feudal societies: wealth is inherently and highly skewed, with a tiny fraction of the population holding the vast majority of resources.

### Definition 7: Pareto's Law

In a stable economy, the number of individuals ( $N_x$ ) having an income greater than or equal to  $x$  is approximately:

$$N_x = \frac{A}{(x+b)^\alpha} \quad \text{for all } x \geq x_0 \quad (22)$$

Where  $A$  is a constant,  $b$  is a constant near zero, and  $\alpha$  is a parameter (typically  $1 < \alpha \leq 2$ ).

Because  $b$  is usually very close to zero, economists simplify this into the **Strong Pareto Law**:  $N_x = Ax^{-\alpha}$ . Most modern income distributions obey the **Weak Pareto Law**, which states that the extreme upper tail of the distribution behaves proportionally to  $x^{-\alpha}$  as  $x \rightarrow \infty$ .

## 12.2 Socioeconomic Rules of Thumb

Pareto's empirical observations gave rise to famous heuristics regarding inequality:

- **The 80/20 Rule (Pareto Principle):** Roughly 80% of consequences come from 20% of causes. In economics, this often manifests as 20% of the population owning 80% of the land or wealth.
- **The Champagne Glass Effect:** Visualized in the 1992 UN Development Report, the global distribution of income resembles a champagne glass, where the richest 20% of the world generates 82.7% of the income, while the poorest 20% generates only 1.4%.

## 12.3 Graphical Test for the Pareto Distribution

To test if empirical data follows a Pareto distribution, statisticians use a graphical log-log plot. Taking the natural logarithm of the Strong Pareto Law yields a linear equation:

$$\ln(N_x) = \ln(A) - \alpha \ln(x) \quad (23)$$

If the data follows Pareto's law, plotting  $\ln(x)$  on the X-axis against  $\ln(N_x)$  on the Y-axis will produce a straight, downward-sloping line with a slope of  $-\alpha$ .

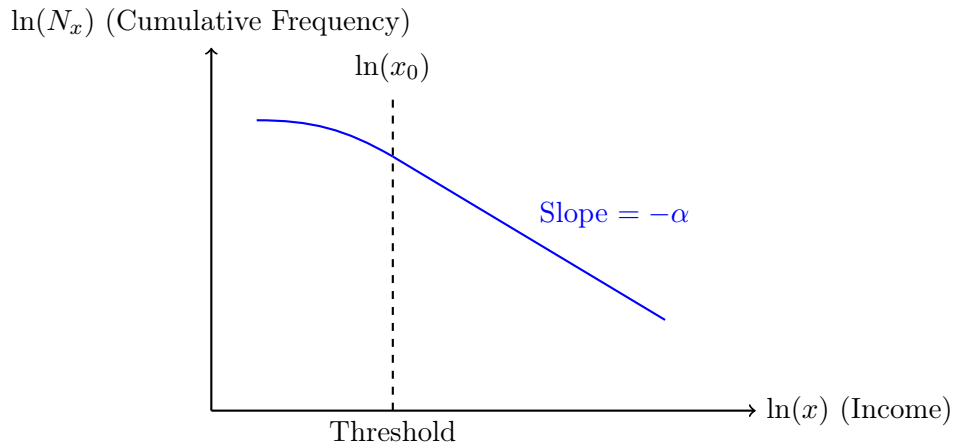


Figure 2: Graphical Test: The linear relationship typically holds only for the upper tail ( $x \geq x_0$ ).

## 12.4 Formal Probability Distribution

Transitioning from empirical observation to pure probability, if we assume  $M$  is the total population and  $N_x$  is the number of people earning at least  $x$ , the probability  $P(X \geq x)$  is  $N_x/M$ . This gives us the formal Pareto Cumulative Distribution Function (CDF) and Probability Density Function (PDF):

### The Pareto Distribution Variables

For a random variable  $X \sim \text{Pareto}(c, \alpha)$  where  $c$  is the minimum possible income ( $x \geq c$ ):

- **CDF:**  $F(x) = 1 - \left(\frac{c}{x}\right)^\alpha$
- **PDF:**  $f(x) = \alpha c^\alpha x^{-\alpha-1}$

## 12.5 Inequality Metrics for the Pareto Distribution

The Lorenz Curve ( $F_1$ ) and the Lorenz Ratio/Gini Coefficient ( $LR$ ) are explicitly defined by the parameter  $\alpha$ :

$$F_1(F) = 1 - (1 - F)^{1-\frac{1}{\alpha}} \quad (24)$$

$$LR = \frac{1}{2\alpha - 1} \quad (25)$$

*Core Intuition:* Inequality varies **inversely** with  $\alpha$ . A higher  $\alpha$  indicates a more egalitarian society, while a lower  $\alpha$  indicates extreme wealth concentration.

## 13 Mathematical Derivations: The Pareto Distribution

Let  $X \sim \text{Pareto}(c, \alpha)$  with PDF  $f(x) = \alpha c^\alpha x^{-\alpha-1}$  for  $x \geq c$  and parameter  $\alpha > 1$ .

### 13.1 Proof 1: Moments of the Pareto Distribution

**The  $r$ -th Moment:**

$$E(X^r) = \int_c^\infty x^r (\alpha c^\alpha x^{-\alpha-1}) dx = \alpha c^\alpha \int_c^\infty x^{r-\alpha-1} dx$$

If  $\alpha > r$ , evaluating the integral gives:  $E(X^r) = \alpha c^\alpha \left[ \frac{x^{r-\alpha}}{r-\alpha} \right]_c^\infty = \frac{\alpha c^r}{\alpha-r}$ . (If  $\alpha \leq r$ , it diverges to  $\infty$ ).

**Mean and Variance:**

Letting  $r = 1$ :  $\mu = E(X) = \frac{\alpha c}{\alpha-1}$ .

Letting  $r = 2$ :  $E(X^2) = \frac{\alpha c^2}{\alpha-2}$ .

$$V(X) = E(X^2) - [E(X)]^2 = \frac{\alpha c^2}{\alpha-2} - \left(\frac{\alpha c}{\alpha-1}\right)^2 = \left(\frac{c}{\alpha-1}\right)^2 \frac{\alpha}{\alpha-2} \quad (\text{for } \alpha > 2).$$

### 13.2 Proof 2: Logarithmic Transformation to Exponential

Let  $Y = \ln(X/c)$ . Therefore,  $X = ce^Y$  and the Jacobian is  $\frac{dx}{dy} = ce^y$ .

$$\begin{aligned} f_Y(y) &= f_X(x) \left| \frac{dx}{dy} \right| \\ f_Y(y) &= [\alpha c^\alpha (ce^y)^{-\alpha-1}] (ce^y) \\ f_Y(y) &= \alpha c^\alpha c^{-\alpha-1} e^{-\alpha y - y} ce^y = \alpha e^{-\alpha y} \end{aligned}$$

Thus,  $Y \sim \text{Exponential}(\alpha)$ .

### 13.3 Proof 3: Geometric and Harmonic Means

**Geometric Mean (GM):**  $\ln(GM) = E(\ln X)$ . Since  $Y = \ln X - \ln c \sim \text{Exp}(\alpha)$ , its expected value is  $E(Y) = 1/\alpha$ .

$$E(\ln X) = \ln c + \frac{1}{\alpha}$$

$$GM = e^{\ln c + 1/\alpha} = c \exp\left(\frac{1}{\alpha}\right)$$

**Harmonic Mean (HM):**  $1/HM = E(X^{-1})$ . Using the  $r$ -th moment formula with  $r = -1$ :

$$E(X^{-1}) = \frac{\alpha c^{-1}}{\alpha - (-1)} = \frac{\alpha}{c(\alpha + 1)}$$

$$HM = c \left(1 + \frac{1}{\alpha}\right)$$

### 13.4 Proof 4: Lorenz Curve and Gini Coefficient

**The Lorenz Curve ( $F_1$ ):**

$$F_1(x) = \frac{1}{\mu} \int_c^x t f(t) dt = \frac{\alpha-1}{\alpha c} \left[ \frac{\alpha c}{\alpha-1} \left(1 - \left(\frac{c}{x}\right)^{\alpha-1}\right) \right] = 1 - \left(\frac{c}{x}\right)^{\alpha-1}$$

Since  $F(x) = 1 - (c/x)^\alpha$ , substituting  $(c/x) = (1 - F)^{1/\alpha}$  yields:

$$F_1(F) = 1 - (1 - F)^{1-1/\alpha}.$$

**The Lorenz Ratio (LR):**

$$LR = 1 - 2 \int_0^1 F_1(F) dF = 1 - 2 \int_0^1 [1 - (1 - F)^{1-1/\alpha}] dF$$

$$LR = 1 - 2 \left(1 - \frac{1}{2-1/\alpha}\right) = 1 - 2 \left(\frac{\alpha-1}{2\alpha-1}\right) = \frac{1}{2\alpha-1}.$$

### 13.5 Proof 5: Continuous Lorenz Ratio Theorem

For IID non-negative variables  $X_1, X_2$  with mean  $\mu$ :

$$E|X_1 - X_2| = 2 \int_0^\infty \int_0^x (x - y) f(y) f(x) dy dx$$

The inner integral simplifies to

$$xF(x) - \mu F_1(x).$$

Integrating over  $f(x) dx$  gives

$$2\mu \left( \int F dF_1 - \int F_1 dF \right).$$

By integration by parts,

$$\int F dF_1 = 1 - \int F_1 dF.$$

Hence,

$$E|X_1 - X_2| = 2\mu \left( 1 - 2 \int_0^1 F_1 dF \right) = 2\mu(LR)$$

Therefore,

$$LR = \frac{E|X_1 - X_2|}{2\mu}.$$

## 14 Estimation of Parameters in the Pareto Distribution

### 14.1 The Primary Challenge: Determining the Cut-off ( $c$ )

The Pareto law typically only holds for the upper tail of an income distribution. Therefore, before estimating the shape parameter ( $\alpha$ ), a statistician must determine the threshold or cut-off point ( $x_0$ , which serves as  $c$ ).

- **Graphical Method:** Plot  $\ln(x)$  against  $\ln(N_x)$ . The threshold  $c$  is subjectively chosen at the point where the chaotic curve stabilizes into a straight, downward-sloping line.
- **Algorithmic Method:** Objective minimization of squared errors often fails because the extreme upper tail can artificially force the cut-off point too high.

Once  $c$  is determined, only the  $n$  observations where  $x \geq c$  are retained and sorted:  $x_1 \leq x_2 \leq \dots \leq x_n$ .

### 14.2 Estimation Using Raw Data

Assuming an i.i.d. sample of exact income values, there are three primary methods to estimate  $\alpha$  and  $c$ :

#### Theorem 9: 1. Method of Moments (MoM)

Equates the sample mean ( $\bar{x}$ ) and sample variance to the theoretical population moments. Using the sample Coefficient of Variation  $C^2(X) = \frac{Var(X)}{[E(X)]^2}$ :

$$\hat{\alpha} = 1 + \sqrt{\frac{1 + C^2(X)}{C^2(X)}}, \quad \hat{c} = \frac{\bar{x}(\hat{\alpha} - 1)}{\hat{\alpha}} \quad (26)$$

*Critical Flaw:* This method fails mathematically if  $\alpha < 2$ , because the theoretical variance of the Pareto distribution approaches infinity.

#### Theorem 10: 2. Modified Method of Moments (Johnson & Kotz)

To avoid the variance issue, this method uses the sample mean ( $\bar{x}$ ) and the first order statistic (the minimum value,  $x_{(1)}$ ):

$$\hat{\alpha} = \frac{n\bar{x} - x_{(1)}}{n(\bar{x} - x_{(1)})}, \quad \hat{c} = \frac{(n\hat{\alpha} - 1)x_{(1)}}{n\hat{\alpha}} \quad (27)$$

#### Theorem 11: 3. Maximum Likelihood Estimation (MLE)

Maximizes the log-likelihood function  $L(c, \alpha)$ . Because the likelihood strictly increases with  $c$ , the MLE for  $c$  is simply the smallest observation:

$$\hat{c} = x_{(1)} = \min(x_1, x_2, \dots, x_n) \quad (28)$$

Setting the partial derivative with respect to  $\alpha$  to zero yields:

$$\hat{\alpha} = \frac{n}{\sum_{i=1}^n \ln\left(\frac{x_i}{\hat{c}}\right)} \quad (29)$$

### 14.3 Estimation Using Grouped Data

When data is binned into frequency classes ( $f_i$ ) with class boundaries ( $x_{i-1}$  to  $x_i$ ), estimation becomes more complex due to the loss of exact values.

- **Modified Method of Moments:** Uses the estimated mean ( $\bar{x}$ ) and a specific low quantile ( $x_P$ , e.g.,  $P = 0.05$ ) to solve the system:

$$\bar{x} = \frac{\hat{\alpha}\hat{c}}{\hat{\alpha} - 1}, \quad x_P = c(1 - P)^{-1/\alpha} \quad (30)$$

- **Maximum Likelihood Estimation:** Treats the grouped frequencies as a Multinomial distribution. The theoretical probability ( $\pi_i$ ) of an observation falling into interval  $i$  is calculated using the CDF:

$$\pi_i = F(x_i) - F(x_{i-1}) = \left(\frac{c}{x_{i-1}}\right)^\alpha - \left(\frac{c}{x_i}\right)^\alpha \quad (31)$$

These probabilities are plugged into the multinomial likelihood function. The system has no closed-form algebraic solution and must be optimized numerically using statistical software.

## 15 The Lognormal Distribution

### 15.1 Introduction and Definition

While the Pareto distribution models the extreme upper tail of an economy (the wealthy), the **Lognormal (LN) Distribution** is frequently used to model the vast middle class. It does *not* satisfy the Weak Pareto Law.

A continuous random variable  $X$  is lognormally distributed if its natural logarithm  $Y = \ln(X)$  follows a Normal distribution.

$$X \sim LN(\mu, \sigma^2) \iff \ln(X) \sim N(\mu, \sigma^2) \quad (32)$$

*Crucial Note:* The parameters  $\mu$  and  $\sigma^2$  are the mean and variance of the **logarithm** of  $X$ , not of  $X$  itself.

### 15.2 Core Properties and Moments

Because  $X$  is the exponentiation of a Normal variable, it is strictly positive and right-skewed.

- **PDF:**  $\lambda(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right)$  for  $x > 0$
- **Mode (Peak):**  $x_m = e^{\mu - \sigma^2}$
- **Median:**  $x_{0.5} = e^\mu$
- **Mean:**  $E(X) = e^{\mu + \frac{1}{2}\sigma^2}$

*Notice the strict mathematical ordering characteristic of right-skewed distributions: Mean > Median > Mode.*

### 15.3 Graphical Test for Lognormality

The  $P$ -th quantile of a Lognormal distribution is given by  $\ln(X_P) = \mu + \sigma t_P$ , where  $t_P$  is the standard Normal deviate (normit) such that  $\Phi(t_P) = P$ . Rearranging this yields a linear equation:

$$t_P = -\frac{\mu}{\sigma} + \frac{1}{\sigma} \ln(X_P) \quad (33)$$

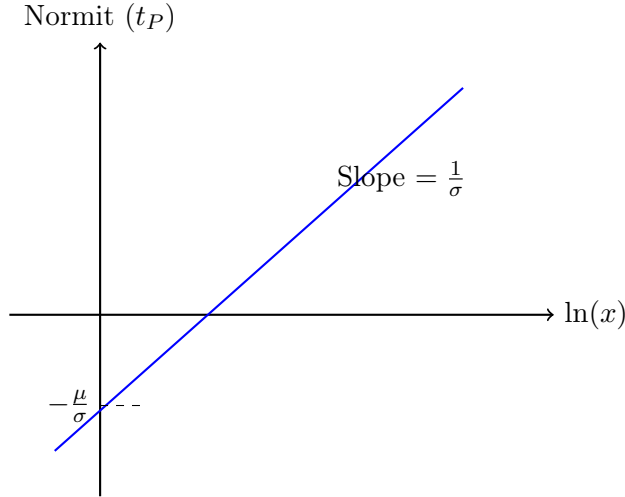


Figure 3: Graphical Test: Plotting the log of income boundaries against their standard Normal quantiles yields a straight line if the data is Lognormal.

## 15.4 Estimation of Parameters

- **Raw Data (MLE):** The Maximum Likelihood Estimators are simply the sample mean and variance of the log-transformed data:

$$\hat{\mu} = \frac{1}{n} \sum \ln(x_j), \quad \hat{\sigma}^2 = \frac{1}{n-1} \sum (\ln(x_j) - \hat{\mu})^2 \quad (34)$$

- **Grouped Data (Method of Quantiles):** Since exact MLE for grouped data requires complex numerical optimization of a multinomial likelihood function, a highly efficient shortcut is to equate two empirical quantiles to the theoretical formula. Using the 27<sup>th</sup> and 73<sup>rd</sup> percentiles for  $\mu$  yields  $\approx 80\%$  asymptotic efficiency.

## 15.5 Advanced Mathematical Proofs

### Theorem 12: The Moment Distribution Theorem

The  $j$ -th moment distribution associated with  $\Lambda(\mu, \sigma^2)$  is also a Lognormal distribution with a shifted mean:  $\Lambda(x|\mu + j\sigma^2, \sigma^2)$ .

*Proof for the 1st Moment ( $j = 1$ ):*

The first moment distribution defines the income share  $F_1(x)$ . Let  $A$  be a normalizing constant.

$$\begin{aligned} d\Lambda_1(x) &= A \cdot x \cdot \lambda(x) dx \\ &= A \cdot e^{\ln x} \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right) dx \\ &= A \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2 - 2\sigma^2 \ln x}{2\sigma^2}\right) dx \end{aligned}$$

Completing the square in the exponent reveals a new mean of  $(\mu + \sigma^2)$ . Thus, the first moment distribution is strictly  $LN(\mu + \sigma^2, \sigma^2)$ .

### Theorem 13: The Lognormal Lorenz Curve

Let  $F = \Phi\left(\frac{\ln x - \mu}{\sigma}\right)$  and  $F_1 = \Phi\left(\frac{\ln x - (\mu + \sigma^2)}{\sigma}\right)$ .

Converting these to standard Normal deviates (normits):

$$t_F = \frac{\ln x - \mu}{\sigma}$$
$$t_{F_1} = \frac{\ln x - \mu - \sigma^2}{\sigma} = \frac{\ln x - \mu}{\sigma} - \sigma$$

Therefore, the exact equation of the Lognormal Lorenz Curve is:

$$t_{F_1} = t_F - \sigma \quad (35)$$

*Implication:* The shape of the inequality curve depends **entirely** on the variance  $\sigma$ , and is completely independent of the mean  $\mu$ . Furthermore, the curve is perfectly symmetrical across the secondary diagonal ( $F_1 + F = 1$ ).

### Theorem 14: The Lognormal Gini Coefficient (LR)

The Lorenz Ratio is exactly  $LR = 2\Phi\left(\frac{\sigma}{\sqrt{2}}\right) - 1$ .

*Proof:*

$LR = 1 - 2 \int_0^1 F_1 dF = 1 - 2P(X_2 \leq X_1)$ , where  $X_1 \sim LN(\mu, \sigma^2)$  and  $X_2 \sim LN(\mu + \sigma^2, \sigma^2)$ .

Let  $Y_1 = \ln(X_1)$  and  $Y_2 = \ln(X_2)$ .

$$P(X_2 \leq X_1) = P(Y_2 - Y_1 \leq 0)$$

Since  $Y_2 - Y_1$  is the difference of two Normal variables, it is distributed as  $N(\sigma^2, 2\sigma^2)$ .

$$P(Y_2 - Y_1 \leq 0) = \Phi\left(\frac{0 - \sigma^2}{\sqrt{2}\sigma^2}\right) = \Phi\left(-\frac{\sigma}{\sqrt{2}}\right) = 1 - \Phi\left(\frac{\sigma}{\sqrt{2}}\right)$$

Substituting this back into the LR equation:

$$LR = 1 - 2 \left[ 1 - \Phi\left(\frac{\sigma}{\sqrt{2}}\right) \right] = 2\Phi\left(\frac{\sigma}{\sqrt{2}}\right) - 1 \quad (36)$$

## 16 Genesis of the Lognormal Distribution

To understand why income often follows a Lognormal distribution, economists use stochastic (random) models. The most famous is Kapteyn's Law of Proportionate Effect, later refined by Kalecki to resolve issues with infinite variance.

## 16.1 The Statistical Foundation

### Theorem 15: Central Limit Theorem (CLT) Corollary

Let  $Y_1, Y_2, \dots, Y_n$  be a sequence of independent random variables taking only positive values. If  $X_i = \ln(Y_i)$ , then under mild conditions, the sum  $\sum X_i$  is asymptotically normally distributed.

*Consequence:* Because  $\ln(\prod Y_i) = \sum \ln(Y_i)$ , the sum of the logs is asymptotically Normal. Therefore, the product of the original variables,  $\prod Y_i$ , is asymptotically **Lognormal**.

## 16.2 Kapteyn's Law of Proportionate Effect (LPE)

The LPE assumes that the change in a variable (like income) during a time interval is a random proportion of its size at the beginning of that period.

Let  $X_0$  be initial income and  $X_t$  be income at time  $t$ . Let  $\epsilon_t$  be a small random shock.

$$X_t - X_{t-1} = \epsilon_t X_{t-1} \implies X_t = X_{t-1}(1 + \epsilon_t) \quad (37)$$

By chaining  $n$  periods together, the final income is the product of all previous shocks:

$$X_n = X_0(1 + \epsilon_1)(1 + \epsilon_2) \dots (1 + \epsilon_n) \quad (38)$$

Taking the natural logarithm of both sides (and letting  $X'_n = \ln(X_n)$  and  $\epsilon'_t = \ln(1 + \epsilon_t)$ ):

$$X'_n = X'_0 + \epsilon'_1 + \epsilon'_2 + \dots + \epsilon'_n \quad (39)$$

Because  $X'_n$  is a sum of independent random variables, by the CLT, it is asymptotically Normal. Thus, the original income  $X_n$  is asymptotically **Lognormal**.

## 16.3 Kalecki's Criticism

Kalecki pointed out a fatal flaw in the basic LPE model. If the random shocks  $\epsilon'_t$  are mutually independent, the variance of the log-income adds up over time:

$$V(X'_n) = V(X'_0) + V(\epsilon'_1) + V(\epsilon'_2) + \dots + V(\epsilon'_n) \quad (40)$$

Because  $V(\ln X)$  is the natural measure of inequality in a Lognormal distribution, this implies that income inequality will grow indefinitely toward infinity as time passes. This contradicts real-world observations where inequality eventually stabilizes.

## 16.4 Kalecki's Modification

To stabilize the variance so that  $V(X'_{n+1}) = V(X'_n)$ , Kalecki argued there must be a **negative correlation** between current income and future proportional growth:

$$Cov(X'_n, \epsilon'_{n+1}) = -\frac{1}{2}V(\epsilon'_{n+1}) < 0 \quad (41)$$

*Economic Intuition:* Government policies (like progressive taxation) retard the growth of high incomes and favor the poor, stabilizing overall inequality.

Kalecki modeled this via linear regression, where  $\eta_t$  is an independent error term:

$$\epsilon'_t = A' - \alpha X'_{t-1} + \eta_t \quad (42)$$

Substituting this back into the growth equation ( $X'_t = X'_{t-1} + \epsilon'_t$ ) yields:

$$X'_t = A' + (1 - \alpha)X'_{t-1} + \eta_t \quad (43)$$

Through recursive substitution,  $X'_t$  becomes a constant plus a weighted sum of uncorrelated errors ( $\eta_t$ ), ensuring it remains asymptotically Normal. Most importantly, the variance becomes a convergent geometric series:

$$V(X'_t) = \sigma_\eta^2 \sum_{r=0}^{\infty} (1 - \alpha)^{2r} \quad (44)$$

Because  $|1 - \alpha| < 1$ , this infinite series converges to a finite constant. Kalecki successfully proved that a Lognormal income distribution can exist without inequality spiraling out of control.

## Practical Assignment 01

**Qn 01:** Calculate (i) Laspeyres', (ii) Paasche's and (iii) Fisher's index numbers from the following data.

| Commodity | Base year price<br>(Rs.) ( $p_0$ ) | Base year<br>quantity ( $q_0$ ) | Current year<br>price (Rs.) ( $p_1$ ) | Current year<br>quantity ( $q_1$ ) |
|-----------|------------------------------------|---------------------------------|---------------------------------------|------------------------------------|
| A         | 10                                 | 30                              | 12                                    | 50                                 |
| B         | 8                                  | 15                              | 10                                    | 25                                 |
| C         | 6                                  | 20                              | 6                                     | 30                                 |
| D         | 4                                  | 10                              | 6                                     | 20                                 |

### Solution to Qn 01

First, we construct the calculation table to find our aggregate sums.

| Commodity                          | $p_0q_0$             | $p_1q_0$             | $p_0q_1$             | $p_1q_1$             |
|------------------------------------|----------------------|----------------------|----------------------|----------------------|
| A                                  | $10 \times 30 = 300$ | $12 \times 30 = 360$ | $10 \times 50 = 500$ | $12 \times 50 = 600$ |
| B                                  | $8 \times 15 = 120$  | $10 \times 15 = 150$ | $8 \times 25 = 200$  | $10 \times 25 = 250$ |
| C                                  | $6 \times 20 = 120$  | $6 \times 20 = 120$  | $6 \times 30 = 180$  | $6 \times 30 = 180$  |
| D                                  | $4 \times 10 = 40$   | $6 \times 10 = 60$   | $4 \times 20 = 80$   | $6 \times 20 = 120$  |
| <b>Total (<math>\Sigma</math>)</b> | <b>580</b>           | <b>690</b>           | <b>960</b>           | <b>1150</b>          |

(i) **Laspeyres' Index ( $L_{01}$ ):**

$$\begin{aligned}
 L_{01} &= \frac{\sum p_1q_0}{\sum p_0q_0} \times 100 \\
 &= \frac{690}{580} \times 100 = 1.18965 \times 100 = \mathbf{118.97}
 \end{aligned}$$

(ii) **Paasche's Index ( $P_{01}$ ):**

$$\begin{aligned}
 P_{01} &= \frac{\sum p_1q_1}{\sum p_0q_1} \times 100 \\
 &= \frac{1150}{960} \times 100 = 1.19791 \times 100 = \mathbf{119.79}
 \end{aligned}$$

(iii) **Fisher's Index ( $F_{01}$ ):**

$$\begin{aligned}
 F_{01} &= \sqrt{L_{01} \times P_{01}} \\
 &= \sqrt{118.965 \times 119.791} = \sqrt{14250.93} = \mathbf{119.38}
 \end{aligned}$$

**Qn 02:** You are given the following information  
Compute price index for 2013 by Fisher's Method.

### Solution to Qn 02

We are given Base Expenditure instead of Base Quantity. Since Expenditure =  $p_0 \times q_0$ , we can find the base quantity by calculating  $q_0 = \text{Expenditure}/p_0$ . Notice that the Expenditure column is already our  $p_0q_0$  sum!

|           | 2010        |                   | 2013        |          |
|-----------|-------------|-------------------|-------------|----------|
| Commodity | Price (Rs.) | Expenditure (Rs.) | Price (Rs.) | Quantity |
| A         | 25          | 1000              | 40          | 50       |
| B         | 22          | 396               | 40          | 30       |
| C         | 54          | 864               | 30          | 44       |
| D         | 20          | 800               | 30          | 45       |
| E         | 18          | 540               | 42          | 15       |

| Commodity          | Derived $q_0$ ( $Exp/p_0$ ) | $p_0q_0$ (Given) | $p_1q_0$              | $p_0q_1$              | $p_1q_1$              |
|--------------------|-----------------------------|------------------|-----------------------|-----------------------|-----------------------|
| A                  | $1000/25 = 40$              | 1000             | $40 \times 40 = 1600$ | $25 \times 50 = 1250$ | $40 \times 50 = 2000$ |
| B                  | $396/22 = 18$               | 396              | $40 \times 18 = 720$  | $22 \times 30 = 660$  | $40 \times 30 = 1200$ |
| C                  | $864/54 = 16$               | 864              | $30 \times 16 = 480$  | $54 \times 44 = 2376$ | $30 \times 44 = 1320$ |
| D                  | $800/20 = 40$               | 800              | $30 \times 40 = 1200$ | $20 \times 45 = 900$  | $30 \times 45 = 1350$ |
| E                  | $540/18 = 30$               | 540              | $42 \times 30 = 1260$ | $18 \times 15 = 270$  | $42 \times 15 = 630$  |
| Total ( $\Sigma$ ) |                             | <b>3600</b>      | <b>5260</b>           | <b>5456</b>           | <b>6500</b>           |

**Fisher's Price Index ( $F_{01}$ ):**

$$\begin{aligned}
 F_{01} &= \sqrt{\frac{\sum p_1q_0}{\sum p_0q_0} \times \frac{\sum p_1q_1}{\sum p_0q_1}} \times 100 \\
 &= \sqrt{\frac{5260}{3600} \times \frac{6500}{5456}} \times 100 \\
 &= \sqrt{1.4611 \times 1.1913} \times 100 \\
 &= \sqrt{1.7406} \times 100 = 1.3193 \times 100 = \mathbf{131.93}
 \end{aligned}$$

**Qn 03:** Calculate Kelly's Price Index from the following data.

| Commodity | Quantity | Price (₹) |              |
|-----------|----------|-----------|--------------|
|           |          | Base year | Current year |
| A         | 10       | 140       | 180          |
| B         | 7        | 400       | 550          |
| C         | 6        | 100       | 250          |
| D         | 8        | 125       | 150          |
| E         | 4        | 200       | 300          |

### Solution to Qn 03

Kelly's Price Index uses a fixed quantity ( $q$ ) as the weight for both the base and current periods. The formula is  $I_K = \frac{\sum p_1q}{\sum p_0q} \times 100$ .

| Commodity                          | $p_0q$                 | $p_1q$                 |
|------------------------------------|------------------------|------------------------|
| A                                  | $140 \times 10 = 1400$ | $180 \times 10 = 1800$ |
| B                                  | $400 \times 7 = 2800$  | $550 \times 7 = 3850$  |
| C                                  | $100 \times 6 = 600$   | $250 \times 6 = 1500$  |
| D                                  | $125 \times 8 = 1000$  | $150 \times 8 = 1200$  |
| E                                  | $200 \times 4 = 800$   | $300 \times 4 = 1200$  |
| <b>Total (<math>\Sigma</math>)</b> | <b>6600</b>            | <b>9550</b>            |

Kelly's Price Index ( $I_K$ ):

$$\begin{aligned}
 I_K &= \frac{\sum p_1q}{\sum p_0q} \times 100 \\
 &= \frac{9550}{6600} \times 100 \\
 &= 1.44696 \times 100 = \mathbf{144.70}
 \end{aligned}$$

## Practical Assignment 02: Advanced Index Numbers

**Qn 01:** Given below are two price index series. Splice them on the base 1974=100. By what per cent did the price of steel rise between 1970 and 1975?

| Year | Old price index for Steel<br>Base (1965=100) | New price index for Steel<br>Base (1974=100) |
|------|----------------------------------------------|----------------------------------------------|
| 1970 | 141.5                                        |                                              |
| 1971 | 163.7                                        |                                              |
| 1972 | 158.2                                        |                                              |
| 1973 | 156.8                                        | 99.8                                         |
| 1974 | 157.1                                        | 100.0                                        |
| 1975 |                                              | 102.3                                        |

### Solution to Qn 01

To splice the old index to the new base (1974=100), we use the overlapping year (1974) to find our splicing factor. Since the old index was 157.1 when the new index was 100.0, we multiply the old series by the factor  $\frac{100}{157.1}$ .

| Year | Old price index<br>(1965=100) | Spliced / New price index<br>(1974=100)            |
|------|-------------------------------|----------------------------------------------------|
| 1970 | 141.5                         | $\frac{100}{157.1} \times 141.5 = \mathbf{90.07}$  |
| 1971 | 163.7                         | $\frac{100}{157.1} \times 163.7 = \mathbf{104.20}$ |
| 1972 | 158.2                         | $\frac{100}{157.1} \times 158.2 = \mathbf{100.70}$ |
| 1973 | 156.8                         | $\frac{100}{157.1} \times 156.8 = \mathbf{99.81}$  |
| 1974 | 157.1                         | <b>100.00</b>                                      |
| 1975 |                               | <b>102.30</b>                                      |

**Percentage Increase (1970 to 1975):**

$$\begin{aligned}\% \text{ Increase} &= \frac{\text{Index}_{1975} - \text{Index}_{1970}}{\text{Index}_{1970}} \times 100 \\ &= \frac{102.30 - 90.07}{90.07} \times 100 = \frac{12.23}{90.07} \times 100 = 0.13578 \times 100 = \mathbf{13.58\%}\end{aligned}$$

**Qn 02:** From the following data compute the Fisher's Price Index Number.

| Commodity | Base Year (0)   |                    | Current Year (1) |                    |
|-----------|-----------------|--------------------|------------------|--------------------|
|           | Price ( $p_0$ ) | Quantity ( $q_0$ ) | Price ( $p_1$ )  | Quantity ( $q_1$ ) |
| A         | 1               | 6                  | 5                | 8                  |
| B         | 2               | 7                  | 4                | 7                  |
| C         | 3               | 8                  | 3                | 6                  |
| D         | 4               | 9                  | 2                | 5                  |

**Solution to Qn 02**

First, we compute the necessary aggregate sums.

| Commodity                          | $p_1q_0$          | $p_0q_0$          | $p_1q_1$          | $p_0q_1$          |
|------------------------------------|-------------------|-------------------|-------------------|-------------------|
| A                                  | $5 \times 6 = 30$ | $1 \times 6 = 6$  | $5 \times 8 = 40$ | $1 \times 8 = 8$  |
| B                                  | $4 \times 7 = 28$ | $2 \times 7 = 14$ | $4 \times 7 = 28$ | $2 \times 7 = 14$ |
| C                                  | $3 \times 8 = 24$ | $3 \times 8 = 24$ | $3 \times 6 = 18$ | $3 \times 6 = 18$ |
| D                                  | $2 \times 9 = 18$ | $4 \times 9 = 36$ | $2 \times 5 = 10$ | $4 \times 5 = 20$ |
| <b>Total (<math>\Sigma</math>)</b> | <b>100</b>        | <b>80</b>         | <b>96</b>         | <b>60</b>         |

**Fisher's Price Index ( $F_{01}$ ):**

$$\begin{aligned}F_{01} &= \sqrt{\frac{\sum p_1q_0}{\sum p_0q_0} \times \frac{\sum p_1q_1}{\sum p_0q_1}} \times 100 \\ &= \sqrt{\frac{100}{80} \times \frac{96}{60}} \times 100 \\ &= \sqrt{1.25 \times 1.60} \times 100 \\ &= \sqrt{2} \times 100 = 1.4142 \times 100 = \mathbf{141.42}\end{aligned}$$

**Qn 03:** Compute (i) Laspeyres', (ii) Paasche's, (iii) Fisher's, (iv) Bowley's and (v) Marshall Edgeworth's index from the following table.

**Solution to Qn 03**

*Note: The given table lists Quantity before Price. We must map our columns carefully.*

**(i) Laspeyres' Price Index ( $L_{01}$ ):**

$$L_{01} = \frac{\sum p_1q_0}{\sum p_0q_0} \times 100 = \frac{505}{425} \times 100 = \mathbf{118.82} \quad (45)$$

| Commodity | Base Year          |                 | Current Year       |                 |
|-----------|--------------------|-----------------|--------------------|-----------------|
|           | Quantity ( $q_0$ ) | Price ( $p_0$ ) | Quantity ( $q_1$ ) | Price ( $p_1$ ) |
| A         | 12                 | 10              | 15                 | 12              |
| B         | 15                 | 7               | 20                 | 5               |
| C         | 24                 | 5               | 20                 | 9               |
| D         | 5                  | 16              | 5                  | 14              |

| Commodity                          | $p_1q_0$             | $p_0q_0$             | $p_1q_1$             | $p_0q_1$             |
|------------------------------------|----------------------|----------------------|----------------------|----------------------|
| A                                  | $12 \times 12 = 144$ | $10 \times 12 = 120$ | $12 \times 15 = 180$ | $10 \times 15 = 150$ |
| B                                  | $5 \times 15 = 75$   | $7 \times 15 = 105$  | $5 \times 20 = 100$  | $7 \times 20 = 140$  |
| C                                  | $9 \times 24 = 216$  | $5 \times 24 = 120$  | $9 \times 20 = 180$  | $5 \times 20 = 100$  |
| D                                  | $14 \times 5 = 70$   | $16 \times 5 = 80$   | $14 \times 5 = 70$   | $16 \times 5 = 80$   |
| <b>Total (<math>\Sigma</math>)</b> | <b>505</b>           | <b>425</b>           | <b>530</b>           | <b>470</b>           |

(ii) Paasche's Price Index ( $P_{01}$ ):

$$P_{01} = \frac{\sum p_1q_1}{\sum p_0q_1} \times 100 = \frac{530}{470} \times 100 = \mathbf{112.77} \quad (46)$$

(iii) Fisher's Price Index ( $F_{01}$ ):

$$F_{01} = \sqrt{L_{01} \times P_{01}} = \sqrt{118.8235 \times 112.7660} = \sqrt{13399.28} = \mathbf{115.76} \quad (47)$$

(iv) Bowley's Price Index ( $B_{01}$ ):

$$B_{01} = \frac{L_{01} + P_{01}}{2} = \frac{118.82 + 112.77}{2} = \mathbf{115.80} \quad (48)$$

(v) Marshall-Edgeworth's Price Index ( $EM_{01}$ ):

$$EM_{01} = \frac{\sum p_1q_0 + \sum p_1q_1}{\sum p_0q_0 + \sum p_0q_1} \times 100 = \frac{505 + 530}{425 + 470} \times 100 = \frac{1035}{895} \times 100 = \mathbf{115.64} \quad (49)$$

## Practical Assignment 03: Algebraic Applications

**Qn 01:** If the ratio between Laspeyres' ( $L$ ) and Paasche's ( $P$ ) index numbers is  $L/P = 28/27$ , find the missing figure in the following table.

| Commodity | Base Year       |                    | Current Year    |                    |
|-----------|-----------------|--------------------|-----------------|--------------------|
|           | Price ( $p_0$ ) | Quantity ( $q_0$ ) | Price ( $p_1$ ) | Quantity ( $q_1$ ) |
| A         | 1               | 10                 | 2               | 5                  |
| B         | 1               | 5                  | $x$             | 2                  |

### Solution to Qn 01

First, we construct our calculation table, keeping  $x$  as an algebraic variable.

Let's express Laspeyres' ( $L$ ) and Paasche's ( $P$ ) indices:

$$L = \frac{\sum p_1q_0}{\sum p_0q_0} \times 100 = \frac{20 + 5x}{15} \times 100 = \frac{4 + x}{3} \times 100$$

$$P = \frac{\sum p_1q_1}{\sum p_0q_1} \times 100 = \frac{10 + 2x}{7} \times 100$$

| Commodity                          | $p_0q_0$           | $p_0q_1$         | $p_1q_0$           | $p_1q_1$          |
|------------------------------------|--------------------|------------------|--------------------|-------------------|
| A                                  | $1 \times 10 = 10$ | $1 \times 5 = 5$ | $2 \times 10 = 20$ | $2 \times 5 = 10$ |
| B                                  | $1 \times 5 = 5$   | $1 \times 2 = 2$ | $x \times 5 = 5x$  | $x \times 2 = 2x$ |
| <b>Total (<math>\Sigma</math>)</b> | <b>15</b>          | <b>7</b>         | <b>20 + 5x</b>     | <b>10 + 2x</b>    |

We are given the ratio  $L/P = 28/27$ . Substituting our expressions (the 100s cancel out):

$$\begin{aligned}\frac{\frac{4+x}{3}}{\frac{10+2x}{7}} &= \frac{28}{27} \\ \frac{7(4+x)}{3(10+2x)} &= \frac{28}{27} \\ \frac{4+x}{10+2x} &= \frac{28}{27} \times \frac{3}{7} \\ \frac{4+x}{10+2x} &= \frac{4}{9}\end{aligned}$$

Cross-multiplying to solve for  $x$ :

$$\begin{aligned}9(4+x) &= 4(10+2x) \\ 36+9x &= 40+8x \\ 9x-8x &= 40-36 \\ x &= 4\end{aligned}$$

*Conclusion: The missing price figure is 4.*

**Qn 02:** Given that  $\sum p_1q_1 = 250$ ,  $\sum p_0q_0 = 150$ , Paasche's index number is 150 and Dorbish and Bowley's index number is 145, find out (i) Fisher's ideal index number and (ii) Marshall Edgeworth index number.

### Solution to Qn 02

To calculate any index, we need all four aggregate sums. We must reverse-engineer the missing sums ( $\sum p_0q_1$  and  $\sum p_1q_0$ ).

**Step 1: Find  $\sum p_0q_1$  using Paasche's Index:**

$$\begin{aligned}P_{01} &= \frac{\sum p_1q_1}{\sum p_0q_1} \times 100 = 150 \\ \frac{250}{\sum p_0q_1} \times 100 &= 150 \implies \sum p_0q_1 = \frac{250 \times 100}{150} = \frac{\mathbf{500}}{\mathbf{3}}\end{aligned}$$

**Step 2: Find Laspeyres' Index using Dorbish-Bowley:**

$$\begin{aligned}B_{01} &= \frac{L_{01} + P_{01}}{2} = 145 \\ \frac{L_{01} + 150}{2} &= 145 \implies L_{01} = 290 - 150 = \mathbf{140}\end{aligned}$$

**Step 3: Find  $\sum p_1q_0$  using Laspeyres' Index:**

$$\begin{aligned}L_{01} &= \frac{\sum p_1q_0}{\sum p_0q_0} \times 100 = 140 \\ \frac{\sum p_1q_0}{150} \times 100 &= 140 \implies \sum p_1q_0 = \frac{140 \times 150}{100} = \mathbf{210}\end{aligned}$$

(i) **Fisher's Ideal Index Number ( $F_{01}$ ):**

$$F_{01} = \sqrt{L_{01} \times P_{01}}$$

$$F_{01} = \sqrt{140 \times 150} = \sqrt{21000} \approx \mathbf{144.91}$$

(ii) **Marshall-Edgeworth Index Number ( $EM_{01}$ ):**

$$EM_{01} = \frac{\sum p_1 q_0 + \sum p_1 q_1}{\sum p_0 q_0 + \sum p_0 q_1} \times 100$$

$$EM_{01} = \frac{210 + 250}{150 + \frac{500}{3}} \times 100 = \frac{460}{\frac{950}{3}} \times 100 = \frac{1380}{950} \times 100 \approx \mathbf{145.26}$$

**Qn 03:** Compute the price index by applying the weighted average of price relatives using Arithmetic Mean, Geometric Mean, and Harmonic Mean.

| Commodities | $p_0$ | $p_1$ | Weights ( $w$ ) |
|-------------|-------|-------|-----------------|
| Sugar       | 10    | 15    | 6               |
| Rice        | 20    | 25    | 10              |
| Milk        | 10    | 14    | 8               |

### Solution to Qn 03

Let the Price Relative be  $R = \frac{p_1}{p_0} \times 100$ . We build a calculation table for the components needed for all three means.

| Item                               | $p_0$ | $p_1$ | $w$       | $R = \frac{p_1}{p_0} \times 100$ | $Rw$        | $\ln(R) \times w$           | $\frac{p_0}{p_1} \times w$  |
|------------------------------------|-------|-------|-----------|----------------------------------|-------------|-----------------------------|-----------------------------|
| Sugar                              | 10    | 15    | 6         | 150                              | 900         | $5.0106 \times 6 = 30.064$  | $(10/15) \times 6 = 4.000$  |
| Rice                               | 20    | 25    | 10        | 125                              | 1250        | $4.8283 \times 10 = 48.283$ | $(20/25) \times 10 = 8.000$ |
| Milk                               | 10    | 14    | 8         | 140                              | 1120        | $4.9416 \times 8 = 39.533$  | $(10/14) \times 8 = 5.714$  |
| <b>Total (<math>\Sigma</math>)</b> |       |       | <b>24</b> |                                  | <b>3270</b> | <b>117.880</b>              | <b>17.714</b>               |

1. **Weighted Arithmetic Mean (AM):**

$$I_{AM} = \frac{\sum Rw}{\sum w} = \frac{3270}{24} = \mathbf{136.25} \quad (50)$$

2. **Weighted Geometric Mean (GM):**

$$\ln(I_{GM}) = \frac{\sum w \ln(R)}{\sum w} = \frac{117.880}{24} = 4.9117$$

$$I_{GM} = e^{4.9117} = \mathbf{135.87}$$

3. **Weighted Harmonic Mean (HM):** Note: Since  $R = \frac{p_1}{p_0} \times 100$ , the reciprocal term  $\frac{w}{R}$  is equivalent to  $\frac{w \cdot p_0}{p_1 \times 100}$ .

$$I_{HM} = \frac{\sum w}{\sum \left( \frac{w \cdot p_0}{p_1} \right)} \times 100 = \frac{24}{17.714} \times 100 = \mathbf{135.48} \quad (51)$$

| Year                        | 2000 | 2001 | 2002 | 2003 | 2004 |
|-----------------------------|------|------|------|------|------|
| Price of item (Rs. per Kg.) | 20   | 25   | 30   | 45   | 63   |

## Practical Assignment 04: Chain and Fixed Base Indices

**Qn 01:** Calculate chain indices and fixed base indices with year 2000 as base from the following data.

### Solution to Qn 01

For a single commodity, the Fixed Base Index ( $P_t/P_0$ ) and the Chain Base Index (multiplying Link Relatives  $P_t/P_{t-1}$ ) will always yield identical results.

| Year | Price | Fixed Base Index ( $I_{0t}$ )    | Link Relatives ( $I_{t-1,t}$ ) | Chain Base Index ( $C_{0t}$ )               |
|------|-------|----------------------------------|--------------------------------|---------------------------------------------|
| 2000 | 20    | $I_{00} = 1.00$                  | —                              | $C_{00} = \mathbf{1.00}$                    |
| 2001 | 25    | $I_{01} = 25/20 = \mathbf{1.25}$ | $I_{01} = 25/20 = 1.25$        | $C_{01} = I_{01} = \mathbf{1.25}$           |
| 2002 | 30    | $I_{02} = 30/20 = \mathbf{1.50}$ | $I_{12} = 30/25 = 1.20$        | $C_{02} = 1.25 \times 1.20 = \mathbf{1.50}$ |
| 2003 | 45    | $I_{03} = 45/20 = \mathbf{2.25}$ | $I_{23} = 45/30 = 1.50$        | $C_{03} = 1.50 \times 1.50 = \mathbf{2.25}$ |
| 2004 | 63    | $I_{04} = 63/20 = \mathbf{3.15}$ | $I_{34} = 63/45 = 1.40$        | $C_{04} = 2.25 \times 1.40 = \mathbf{3.15}$ |

**Qn 02:** Calculate chain indices and fixed base indices using Laspeyres' formula with year 2010 as base from the following data.

| Comm. | 2010 (0) |       | 2011 (1) |       | 2012 (2) |       | 2013 (3) |       |
|-------|----------|-------|----------|-------|----------|-------|----------|-------|
|       | $p_0$    | $q_0$ | $p_1$    | $q_1$ | $p_2$    | $q_2$ | $p_3$    | $q_3$ |
| A     | 25       | 50    | 30       | 55    | 35       | 55    | 40       | 50    |
| B     | 20       | 30    | 20       | 35    | 30       | 35    | 40       | 30    |
| C     | 55       | 45    | 60       | 40    | 60       | 40    | 65       | 35    |
| D     | 20       | 45    | 25       | 40    | 30       | 35    | 30       | 45    |
| E     | 20       | 15    | 25       | 20    | 30       | 20    | 40       | 15    |

### Solution to Qn 02

First, we compute the necessary aggregate cross-products. Fixed base indices require base year quantities ( $q_0$ ), while Chain base link relatives require the previous year's quantities.

1. Fixed Base and Chain Base Indices (Ratios):
2. Final Indices (Expressed as Percentages):

## Practical Assignment 05: Lorenz Curve and Lorenz Ratio

**Qn 01:** Draw the Lorenz Curve and calculate the Lorenz Ratio by (i) graphical and by (ii) numerical methods, from the following size distribution of Per-Capita Expenditure (PCE): All India rural plus urban: December 1955 – May 1956.

| Comm.            | $p_0q_0$    | $p_1q_0$    | $p_2q_0$    | $p_3q_0$    | $p_1q_1$    | $p_2q_1$    | $p_2q_2$    | $p_3q_2$    |
|------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| A                | 1250        | 1500        | 1750        | 2000        | 1650        | 1925        | 1925        | 2200        |
| B                | 600         | 600         | 900         | 1200        | 700         | 1050        | 1050        | 1400        |
| C                | 2475        | 2700        | 2700        | 2925        | 2400        | 2400        | 2400        | 2600        |
| D                | 900         | 1125        | 1350        | 1350        | 1000        | 1200        | 1050        | 1050        |
| E                | 300         | 375         | 450         | 600         | 500         | 600         | 600         | 800         |
| Sum ( $\Sigma$ ) | <b>5525</b> | <b>6300</b> | <b>7150</b> | <b>8075</b> | <b>6250</b> | <b>7175</b> | <b>7025</b> | <b>8050</b> |

| Year | Fixed Base Index ( $I_{0t}$ )                  | Link Relatives ( $I_{t-1,t}$ )        | Chain Base Index ( $C_{0t}$ )                     |
|------|------------------------------------------------|---------------------------------------|---------------------------------------------------|
| 2010 | $I_{00} = 1.0000$                              | —                                     | $C_{00} = \mathbf{1.0000}$                        |
| 2011 | $I_{01} = \frac{6300}{5525} = \mathbf{1.1403}$ | $I_{01} = \frac{6300}{5525} = 1.1403$ | $C_{01} = I_{01} = \mathbf{1.1403}$               |
| 2012 | $I_{02} = \frac{7150}{5525} = \mathbf{1.2941}$ | $I_{12} = \frac{7175}{6250} = 1.1480$ | $C_{02} = 1.1403 \times 1.1480 = \mathbf{1.3091}$ |
| 2013 | $I_{03} = \frac{8075}{5525} = \mathbf{1.4615}$ | $I_{23} = \frac{8050}{7025} = 1.1459$ | $C_{03} = 1.3091 \times 1.1459 = \mathbf{1.5001}$ |

| Year     | Fixed Base Index | Chain Base Index |
|----------|------------------|------------------|
| 2010 (0) | 100.00           | 100.00           |
| 2011 (1) | 114.03           | 114.03           |
| 2012 (2) | 129.41           | 130.91           |
| 2013 (3) | 146.15           | 150.01           |

| PCE in Rs. ( $x$ ) | Percentage of persons ( $f_i\%$ ) | Average PCE ( $\bar{x}_i$ ) |
|--------------------|-----------------------------------|-----------------------------|
| 0 – 8              | 14.01                             | 6.26                        |
| 8 – 11             | 17.31                             | 9.41                        |
| 11 – 13            | 11.74                             | 11.98                       |
| 13 – 15            | 9.14                              | 13.96                       |
| 15 – 18            | 11.55                             | 16.49                       |
| 18 – 21            | 8.21                              | 19.54                       |
| 21 – 24            | 7.04                              | 22.51                       |
| 24 – 28            | 5.53                              | 25.79                       |
| 28 – 34            | 5.17                              | 30.68                       |
| 34 – 43            | 4.85                              | 37.72                       |
| 43 – 55            | 2.59                              | 47.24                       |
| 55 – $\infty$      | 2.86                              | 83.29                       |
| <b>All</b>         | <b>100.00</b>                     | <b>18.74 (Approx)</b>       |

### Solution (Numerical Method)

First, we convert the percentages into relative frequencies ( $p_i$ ) and calculate the exact mean. Mean Income ( $\bar{x}$ ) =  $\sum(p_i \times \bar{x}_i) \approx \mathbf{18.7292}$ .

Next, we calculate the income share  $q_i = (p_i \times \bar{x}_i)/\bar{x}$ , and build our cumulative columns ( $P_i$  and  $Q_i$ ).

| Class<br>$x$                       | Rel. Freq.<br>$(p_i)$ | Cum. Freq.<br>$(P_i)$ | Avg PCE<br>$(\bar{x}_i)$ | Inc. Share<br>$(q_i)$ | Cum. Share<br>$(Q_i)$ | Trapezium Sum<br>$(Q_i + Q_{i-1})$ | Weighted Area<br>$p_i(Q_i + Q_{i-1})$ |
|------------------------------------|-----------------------|-----------------------|--------------------------|-----------------------|-----------------------|------------------------------------|---------------------------------------|
| 0 – 8                              | 0.1401                | 0.1401                | 6.26                     | 0.0468                | 0.0468                | 0.0468                             | 0.0066                                |
| 8 – 11                             | 0.1731                | 0.3132                | 9.41                     | 0.0870                | 0.1338                | 0.1806                             | 0.0313                                |
| 11 – 13                            | 0.1174                | 0.4306                | 11.98                    | 0.0751                | 0.2089                | 0.3427                             | 0.0402                                |
| 13 – 15                            | 0.0914                | 0.5220                | 13.96                    | 0.0681                | 0.2770                | 0.4859                             | 0.0444                                |
| 15 – 18                            | 0.1155                | 0.6375                | 16.49                    | 0.1017                | 0.3787                | 0.6557                             | 0.0757                                |
| 18 – 21                            | 0.0821                | 0.7196                | 19.54                    | 0.0857                | 0.4644                | 0.8431                             | 0.0692                                |
| 21 – 24                            | 0.0704                | 0.7900                | 22.51                    | 0.0846                | 0.5490                | 1.0133                             | 0.0713                                |
| 24 – 28                            | 0.0553                | 0.8453                | 25.79                    | 0.0761                | 0.6251                | 1.1741                             | 0.0649                                |
| 28 – 34                            | 0.0517                | 0.8970                | 30.68                    | 0.0847                | 0.7098                | 1.3349                             | 0.0690                                |
| 34 – 43                            | 0.0485                | 0.9455                | 37.72                    | 0.0977                | 0.8075                | 1.5173                             | 0.0736                                |
| 43 – 55                            | 0.0259                | 0.9714                | 47.24                    | 0.0653                | 0.8728                | 1.6803                             | 0.0435                                |
| 55 – $\infty$                      | 0.0286                | 1.0000                | 83.29                    | 0.1272                | 1.0000                | 1.8728                             | 0.0536                                |
| <b>Total (<math>\Sigma</math>)</b> | <b>1.0000</b>         |                       |                          | <b>1.0000</b>         |                       |                                    | <b><math>\Sigma = 0.6434</math></b>   |

**Lorenz Ratio (LR):**

$$LR = 1 - \sum_{i=1}^K p_i(Q_i + Q_{i-1}) = 1 - 0.6434 = \mathbf{0.3566} \quad (52)$$

### Solution (Graphical Method)

To draw the Lorenz Curve, we plot the Cumulative Population points ( $P_i$ ) on the X-axis against the Cumulative Income points ( $Q_i$ ) on the Y-axis.

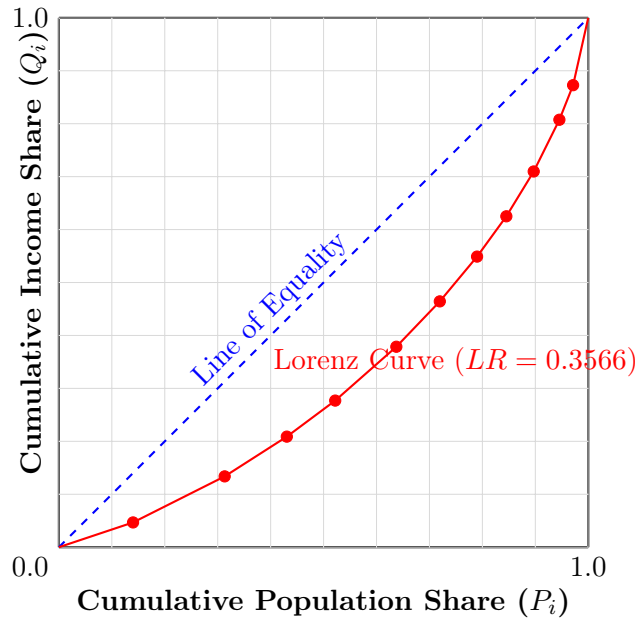


Figure 4: Lorenz Curve for All India PCE (1955-1956)

## MID SEM PART-A

### Question 1 [20 Marks]

**Describe the different steps towards construction of an Index Number.**

**Answer:** The construction of a reliable Index Number involves several systematic steps, as any error in the foundation will compound into the final calculation. The major steps are:

1. **Purpose and Object of the Index:** The exact purpose must be clearly defined (e.g., measuring retail inflation vs. wholesale prices). This dictates what data will be collected and what formulas will be used.
2. **Selection of the Base Period:** The base period must be a "normal" period, free from economic anomalies like wars, famines, or hyperinflation. It should also not be too distant in the past, as consumption patterns change over time.
3. **Selection of Commodities (The Basket):** It is impossible to include every commodity in an economy. A representative sample of goods must be selected based on the consumption habits of the target demographic.
4. **Collection of Prices:** Data must be collected from reliable, standardized sources. A decision must be made between using wholesale prices (which are easier to standardize) or retail prices (which are more relevant to consumers but vary by location).
5. **Selection of Weights:** Not all commodities have the same economic importance (e.g., wheat is more important than salt). Explicit weights (quantities or value shares) must be assigned to reflect the relative importance of each item in the basket.
6. **Selection of Average:** An appropriate mathematical average must be chosen to combine the price relatives. The Geometric Mean is theoretically superior as it is less affected by extreme values and satisfies the time reversal test, but the Arithmetic Mean is often used for computational simplicity.
7. **Selection of Formula:** The final mathematical formula (Laspeyres, Paasche, Fisher, Marshall-Edgeworth, etc.) must be chosen based on the availability of base vs. current year quantity data and the specific requirements of the index.

### Question 2 [20 Marks]

- (i) State and explain the three types of errors in an Index Number formula. [10]  
(ii) What are the different uses of Index Numbers. Explain. [10]

**Answer (i): Three Types of Errors**

1. **Formula Error:** This error arises from the mathematical bias inherent in the chosen formula. For example, Laspeyres' index tends to have an upward bias (overestimating inflation) because it ignores the substitution effect (consumers buying less of goods that become expensive). Paasche's index has a downward bias for the opposite reason.
2. **Sampling Error:** Because an index number uses a representative "basket" of goods rather than every single commodity in the economy, it is subject to statistical sampling error. If the selected basket does not perfectly represent true market consumption, the index will be skewed.

3. **Homogeneity (Quality) Error:** Over time, the quality of a commodity might drastically improve (e.g., smartphones or cars) while its price increases. An index number formula often captures the price increase as pure inflation, failing to account for the fact that the consumer is actually getting a better, higher-quality product.

**Answer (ii): Uses of Index Numbers**

1. **Economic Barometers:** Index numbers act as the pulse of the economy, measuring changes in general price levels, industrial production, and agricultural output over time.
2. **Formulation of Policies:** Governments use indices (like the Consumer Price Index) to formulate fiscal and monetary policies, such as setting interest rates, taxation levels, and price controls.
3. **Wage Adjustments:** Index numbers are directly used to calculate Dearness Allowance (DA) and adjust the salaries of employees to ensure their real purchasing power is not eroded by inflation.
4. **Deflating Value Series:** They are used to adjust nominal economic variables into "real" terms. For example, dividing Nominal GDP by a price index yields Real GDP, stripping away the illusion of growth caused purely by inflation.

**Question 3 [20 Marks]**

(i) Define size distribution. Give examples of some size distributions. Explain why and how share of income is so important in explaining inequality of distribution. [10]

(ii) Define and explain LC and LR. What are the properties of a LC? [10]

**Answer (i): Size Distribution and Income Share** A size distribution is a probability distribution used to model non-negative, continuous economic variables possessing a finite expectation. Examples include the distribution of income, wealth, land holdings, and firm sizes.

For a distribution to be economically valid, the **transfer property** must hold, meaning it is logically possible to transfer a quantity of the variable from one entity to another (e.g., money can be transferred; human height cannot). **Share of income** is critical in explaining inequality because inequality is a relative concept. Absolute wealth values do not show disparity; we must calculate what percentage of the total economic "pie" is held by a specific percentage of the population (e.g., the bottom 40% holding 20% of aggregate income).

**Answer (ii): Lorenz Curve (LC) and Lorenz Ratio (LR)** Let  $f(x)$  be the probability density function of income with mean  $\mu$ .

- $F(x) = \int_0^x f(t)dt$  represents the cumulative population share.
- $F_1(x) = \frac{1}{\mu} \int_0^x tf(t)dt$  represents the cumulative income share.

The **Lorenz Curve (LC)** is the parametric graph of  $F_1$  against  $F$ . The **Lorenz Ratio (LR)** (Gini Coefficient) quantifies inequality as exactly twice the Area of Concentration between the egalitarian line and the LC:  $LR = 1 - 2 \int_0^1 F_1 dF$ .

**Properties of the Lorenz Curve:**

1. It begins at  $(0, 0)$  and ends at  $(1, 1)$ .
2. It is strictly monotonically increasing ( $\frac{dF_1}{dF} > 0$ ).
3. It is strictly convex from below ( $\frac{d^2 F_1}{dF^2} > 0$ ), sagging below the Line of Equality.

## Question 4 [20 Marks]

- (i) State Pareto law. Write down the evidence for and against Pareto law. [10]  
(ii) Derive Pareto density from Pareto law. Prove the  $r$ -th moment and variance. [10]

**Answer (i): Pareto's Law and Evidence** Pareto's Law states that the number of individuals ( $N_x$ ) with an income greater than or equal to  $x$  follows the highly skewed formula:  $N_x = \frac{A}{(x+b)^\alpha}$  for  $x \geq x_0$ .

- **Evidence For:** Proponents argue the law holds for the upper tail of almost all non-socialist countries. It is likened to "Lotka's inverse square law of scientific productivity," suggesting that extreme wealth concentration is a natural reflection of skewed human talent.
- **Evidence Against:** Critics note that socialist nations with strict income ceilings violate this law. Furthermore, the extreme right tail is driven primarily by inherited *property income*, not talent-based *wage income*.

**Answer (ii): Mathematical Derivation** The probability of income  $\geq x$  is  $1 - F(x) = \left(\frac{x}{c}\right)^{-\alpha}$  for  $x \geq c$ . Taking the derivative yields the **Pareto Density Function**:

$$f(x) = \frac{d}{dx} \left[ 1 - \left(\frac{x}{c}\right)^{-\alpha} \right] = \alpha c^\alpha x^{-\alpha-1} \quad (x \geq c) \quad (53)$$

**Proof of the  $r$ -th Moment ( $E(X^r)$ ):**

$$E(X^r) = \int_c^\infty x^r (\alpha c^\alpha x^{-\alpha-1}) dx = \alpha c^\alpha \int_c^\infty x^{r-\alpha-1} dx.$$

If  $\alpha \leq r$ , the integral diverges. If  $\alpha > r$ , the power rule gives:

$$E(X^r) = \begin{cases} \infty & \text{if } \alpha \leq r \\ \frac{\alpha c^r}{\alpha-r} & \text{if } \alpha > r \end{cases} \quad (54)$$

**Proof of Variance ( $V(X)$ ):**

$$V(X) = E(X^2) - [E(X)]^2. \text{ If } \alpha \in (1, 2], E(X^2) = \infty. \text{ If } \alpha > 2: V(X) = \frac{\alpha c^2}{\alpha-2} - \left(\frac{\alpha c}{\alpha-1}\right)^2 = \left(\frac{c}{\alpha-1}\right)^2 \frac{\alpha}{\alpha-2}.$$

$$V(X) = \begin{cases} \infty & \text{if } \alpha \in (1, 2] \\ \left(\frac{c}{\alpha-1}\right)^2 \frac{\alpha}{\alpha-2} & \text{if } \alpha > 2 \end{cases} \quad (55)$$

## Question 5 [10 Marks]

**Write a short note on any one of the following.** (All three provided for complete study reference)

**(a) True Cost of Living Index Number:** The True Cost of Living Index (TCLI), such as the Konus index, is a theoretical microeconomic construct. Unlike standard statistical indices (like Laspeyres) which track the cost of a *fixed basket* of goods, the TCLI measures the ratio of the minimum expenditure required in the current period to the minimum expenditure required in the base period, while keeping the consumer's *utility* (or standard of living) constant. It perfectly accounts for the substitution effect, as it assumes rational consumers will alter their basket to buy cheaper alternatives while maintaining the same level of satisfaction.

**(b) Graphical Test for Pareto Distribution:** To test if empirical data follows a Pareto distribution, we use a log-log plot based on the Strong Pareto Law ( $N_x = Ax^{-\alpha}$ ). Taking the natural logarithm yields:  $\ln(N_x) = \ln(A) - \alpha \ln(x)$ . Plotting  $\ln(x)$  on the X-axis against  $\ln(N_x)$

on the Y-axis should theoretically produce a straight line. In reality, the lower/middle income data will appear curved, but the extreme upper tail (above threshold  $c$ ) will stabilize into a perfectly straight line with a downward slope of  $-\alpha$ .

**(c) Estimation of parameters of LN distribution:** Estimating the parameters ( $\mu$  and  $\sigma^2$ ) of a Lognormal distribution depends on the data type. Because taking the logarithm of LN data transforms it into a standard Normal distribution, for **Raw Data**, the Maximum Likelihood Estimators (MLE) are simply the arithmetic mean and variance of the log-transformed values:  $\hat{\mu} = \frac{1}{n} \sum \ln(x_j)$  and  $\hat{\sigma}^2 = \frac{1}{n-1} \sum (\ln(x_j) - \hat{\mu})^2$ . For **Grouped Data**, exact MLE requires complex numerical optimization of a multinomial function. Therefore, a highly efficient shortcut is the **Method of Quantiles**, where equating two empirical quantiles (e.g., 27<sup>th</sup> and 73<sup>rd</sup>) to the theoretical formula  $\ln(X_P) = \mu + \sigma t_P$  allows for a rapid algebraic solution with roughly 80% efficiency.

## Part B: Practical Solutions

### Question (A) [20 Marks]

Compute (i) Laspeyres', (ii) Paasche's, (iii) Fisher's, (iv) Bowley's and (v) Marshall Edgeworth's index numbers from the following data on monthly consumption of a household.

| Commodity | Base year price<br>(Rs.) ( $p_0$ ) | Base year<br>quantity ( $q_0$ ) | Current year<br>price (Rs.) ( $p_1$ ) | Current year<br>quantity ( $q_1$ ) |
|-----------|------------------------------------|---------------------------------|---------------------------------------|------------------------------------|
| Rice      | 60                                 | 20                              | 60                                    | 25                                 |
| Wheat     | 30                                 | 15                              | 35                                    | 10                                 |
| Fish      | 400                                | 20                              | 450                                   | 25                                 |
| Milk      | 60                                 | 10                              | 65                                    | 12                                 |

**Solution:** First, we construct a table to calculate the four aggregate cross-products required for the indices.

| Commodity                          | $p_0q_0$               | $p_1q_0$               | $p_0q_1$                | $p_1q_1$                |
|------------------------------------|------------------------|------------------------|-------------------------|-------------------------|
| Rice                               | $60 \times 20 = 1200$  | $60 \times 20 = 1200$  | $60 \times 25 = 1500$   | $60 \times 25 = 1500$   |
| Wheat                              | $30 \times 15 = 450$   | $35 \times 15 = 525$   | $30 \times 10 = 300$    | $35 \times 10 = 350$    |
| Fish                               | $400 \times 20 = 8000$ | $450 \times 20 = 9000$ | $400 \times 25 = 10000$ | $450 \times 25 = 11250$ |
| Milk                               | $60 \times 10 = 600$   | $65 \times 10 = 650$   | $60 \times 12 = 720$    | $65 \times 12 = 780$    |
| <b>Total (<math>\Sigma</math>)</b> | <b>10250</b>           | <b>11375</b>           | <b>12520</b>            | <b>13880</b>            |

(i) Laspeyres' Index ( $L_{01}$ ):

$$L_{01} = \frac{\sum p_1q_0}{\sum p_0q_0} \times 100 = \frac{11375}{10250} \times 100 = \mathbf{110.98} \quad (56)$$

(ii) Paasche's Index ( $P_{01}$ ):

$$P_{01} = \frac{\sum p_1q_1}{\sum p_0q_1} \times 100 = \frac{13880}{12520} \times 100 = \mathbf{110.86} \quad (57)$$

(iii) Fisher's Index ( $F_{01}$ ):

$$F_{01} = \sqrt{L_{01} \times P_{01}} = \sqrt{110.9756 \times 110.8626} = \mathbf{110.92} \quad (58)$$

(iv) Bowley's Index ( $B_{01}$ ):

$$B_{01} = \frac{L_{01} + P_{01}}{2} = \frac{110.9756 + 110.8626}{2} = \mathbf{110.92} \quad (59)$$

(v) Marshall-Edgeworth Index ( $EM_{01}$ ):

$$EM_{01} = \frac{\sum p_1q_0 + \sum p_1q_1}{\sum p_0q_0 + \sum p_0q_1} \times 100 = \frac{11375 + 13880}{10250 + 12520} \times 100 = \frac{25255}{22770} \times 100 = \mathbf{110.91} \quad (60)$$

| Year | Old Index (1965=100) | New Index (1974=100) | Spliced Index (1974=100)                       |
|------|----------------------|----------------------|------------------------------------------------|
| 1970 | 155.65               |                      | $155.65 \times (100/172.81) = \mathbf{90.07}$  |
| 1971 | 180.07               |                      | $180.07 \times (100/172.81) = \mathbf{104.20}$ |
| 1972 | 174.02               |                      | $174.02 \times (100/172.81) = \mathbf{100.70}$ |
| 1973 | 172.48               | 99.81                | $172.48 \times (100/172.81) = \mathbf{99.81}$  |
| 1974 | 172.81               | 100.00               | <b>100.00</b>                                  |
| 1975 |                      | 102.30               | <b>102.30</b>                                  |

### Question (B)

Given below are two price index series. Splice them on the base 1974=100. By what per cent did the price of steel rise between 1970 and 1975?

**Solution:** We use the overlapping year (1974) to determine the splicing factor:  $\frac{\text{New Index}}{\text{Old Index}} = \frac{100.00}{172.81}$ .

**Percentage Rise (1970 to 1975):**

$$\% \text{ Rise} = \frac{102.30 - 90.07}{90.07} \times 100 = \frac{12.23}{90.07} \times 100 = \mathbf{13.58\%} \quad (61)$$

### Question 2 [20 Marks]

Calculate chain indices and fixed base indices using Laspeyres' formula with year 2010 as base from the following data.

**Solution:** Fixed Base Laspeyres uses base year weights ( $q_0$ ). Chain Base Laspeyres Link Relatives use the previous year's weights ( $q_{t-1}$ ). We calculate all necessary cross-products below.

| Comm.              | $p_0q_0$       | $p_1q_0$       | $p_2q_0$       | $p_3q_0$       | $p_1q_1$       | $p_2q_1$       | $p_2q_2$       | $p_3q_2$       |
|--------------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| A                  | $10(20) = 200$ | $10(20) = 200$ | $12(20) = 240$ | $15(20) = 300$ | $10(20) = 200$ | $12(20) = 240$ | $12(20) = 240$ | $15(20) = 300$ |
| B                  | $10(15) = 150$ | $12(15) = 180$ | $15(15) = 225$ | $15(15) = 225$ | $12(15) = 180$ | $15(15) = 225$ | $15(18) = 270$ | $15(18) = 270$ |
| C                  | $20(20) = 400$ | $20(20) = 400$ | $20(20) = 400$ | $22(20) = 440$ | $20(20) = 400$ | $20(20) = 400$ | $20(20) = 400$ | $22(20) = 440$ |
| D                  | $10(20) = 200$ | $10(20) = 200$ | $12(20) = 240$ | $12(20) = 240$ | $10(22) = 220$ | $12(22) = 264$ | $12(22) = 264$ | $12(22) = 264$ |
| E                  | $5(15) = 75$   | $6(15) = 90$   | $7(15) = 105$  | $8(15) = 120$  | $6(20) = 120$  | $7(20) = 140$  | $7(20) = 140$  | $8(20) = 160$  |
| Total ( $\Sigma$ ) | <b>1025</b>    | <b>1070</b>    | <b>1210</b>    | <b>1325</b>    | <b>1120</b>    | <b>1269</b>    | <b>1314</b>    | <b>1434</b>    |

#### Final Indices Calculation:

| Year | Fixed Base ( $I_{0t} = \frac{\sum p_t q_0}{\sum p_0 q_0} \times 100$ ) | Link Relatives ( $LR_t = \frac{\sum p_t q_{t-1}}{\sum p_{t-1} q_{t-1}}$ ) | Chain Base ( $C_t = C_{t-1} \times LR_t$ ) |
|------|------------------------------------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------|
| 2010 | $1.00 \times 100 = \mathbf{100.00}$                                    | —                                                                         | <b>100.00</b>                              |
| 2011 | $(1070/1025) \times 100 = \mathbf{104.39}$                             | $1070/1025 = 1.0439$                                                      | $100.00 \times 1.0439 = \mathbf{104.39}$   |
| 2012 | $(1210/1025) \times 100 = \mathbf{118.05}$                             | $1269/1120 = 1.1330$                                                      | $104.39 \times 1.1330 = \mathbf{118.27}$   |
| 2013 | $(1325/1025) \times 100 = \mathbf{129.27}$                             | $1434/1314 = 1.0913$                                                      | $118.27 \times 1.0913 = \mathbf{129.07}$   |

### Question 3 [10 Marks]

If the ratio between Laspeyres' (L) and Paasche's (P) index numbers is  $L/P = 176/175$ , find the missing figure in the following table and hence find L and P.

**Solution:** Let  $x$  be the missing price ( $p_1$ ) for Commodity B. We calculate the aggregate sums algebraically:

- $\sum p_0q_0 = (5 \times 4) + (6 \times 5) = 20 + 30 = \mathbf{50}$
- $\sum p_1q_0 = (6 \times 4) + (x \times 5) = \mathbf{24 + 5x}$
- $\sum p_0q_1 = (5 \times 5) + (6 \times 5) = 25 + 30 = \mathbf{55}$
- $\sum p_1q_1 = (6 \times 5) + (x \times 5) = \mathbf{30 + 5x}$

Laspeyres' index (without the 100 multiplier, as it cancels in the ratio) is  $L = \frac{24+5x}{50}$ .  
 Paasche's index is  $P = \frac{30+5x}{55}$ .  
 Given the ratio  $L/P = 176/175$ :

$$\begin{aligned} \frac{\frac{24+5x}{50}}{\frac{30+5x}{55}} &= \frac{176}{175} \\ \frac{24+5x}{50} \times \frac{55}{30+5x} &= \frac{176}{175} \\ \frac{11(24+5x)}{10(30+5x)} &= \frac{176}{175} \end{aligned}$$

Divide both sides by 11 and multiply both sides by 10:

$$\frac{24+5x}{30+5x} = \frac{16}{175} \times 10 = \frac{160}{175} = \frac{32}{35}$$

Cross-multiply to solve for  $x$ :

$$\begin{aligned} 35(24+5x) &= 32(30+5x) \\ 840+175x &= 960+160x \\ 15x &= 120 \\ x &= \mathbf{8} \end{aligned}$$

*The missing price for Commodity B is 8.*

Finally, plug  $x = 8$  back into the index formulas to find L and P:

$$\begin{aligned} L &= \frac{24+5(8)}{50} \times 100 = \frac{64}{50} \times 100 = \mathbf{128.00} \\ P &= \frac{30+5(8)}{55} \times 100 = \frac{70}{55} \times 100 = \mathbf{127.27} \end{aligned}$$

## 17 Measures of Inequality

### 17.1 Size Distribution & Properties

A **Size Distribution** is a non-negative continuous probability distribution with a finite expectation (e.g., income, wealth, land holdings).

- **Transfer Property:** For a variable to be a valid size distribution for inequality measurement, it must be physically possible to transfer a quantity of it from one person to another (e.g., income can be transferred; human height cannot).

### Theorem 16: Desirable Axioms for Inequality Measures

Let  $I(y_1, y_2, \dots, y_n)$  be an inequality measure. It must satisfy:

1. **Scale Independence:**  $I(cy_1, cy_2, \dots, cy_n) = I(y_1, y_2, \dots, y_n)$  for  $c > 0$ .
2. **Pigou-Dalton Principle of Transfers:** Transferring income from a richer person to a poorer person (without reversing their ranks) strictly decreases inequality.
3. **Principle of Diminishing Transfers:** A transfer between two poor people reduces overall inequality more than the exact same transfer between two rich people.

## 17.2 Positive Measures of Inequality

Positive measures are based purely on statistical dispersion.

### 1. Measures of Relative Dispersion:

- **Relative Mean Deviation (RMD):**  $RMD = \frac{\frac{1}{n} \sum_{i=1}^n |y_i - \bar{y}|}{\bar{y}}$
- **Coefficient of Variation (CV):**  $CV = \frac{SD}{AM} = \frac{\sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2}}{\bar{y}}$
- **Standard Deviation of Logarithms (SDL):**  $SDL = \sqrt{\frac{1}{n} \sum_{i=1}^n (\ln(\frac{y_i}{G}))^2}$   
(where  $G$  is the geometric mean).

### Definition 8: Gini Concentration Coefficient (Lorenz Ratio)

An average of absolute differences between all pairs of incomes. For sorted incomes  $y_1 \leq y_2 \leq \dots \leq y_n$ :

$$LR = 1 - \frac{1}{n^2 \bar{y}} \sum_{i=1}^n \{2(n-i) + 1\} y_i \quad (62)$$

Equivalent to:  $LR = \frac{1}{2n^2 \bar{y}} \sum_{i=1}^n \sum_{j=1}^n |y_i - y_j|$

### 2. Entropy and EF Measures:

- **Theil's Entropy Measure (T):** Let income share  $x_i = \frac{y_i}{n\bar{y}}$ .  
 $T = \sum_{i=1}^n x_i \ln(nx_i)$ . (Limits: 0 for perfect equality,  $\ln(n)$  for perfect inequality).
- **Generalized Entropy (GE):**  $GE(\alpha) = \frac{1}{\alpha(\alpha-1)} \left[ \frac{1}{n} \sum_{i=1}^n \left( \frac{y_i}{\bar{y}} \right)^\alpha - 1 \right]$  for  $\alpha \geq 0$ .
- **Elteto-Frigyes (EF):** Let  $m$  = overall mean,  $m_1$  = mean of incomes  $< m$ , and  $m_2$  = mean of incomes  $> m$ . The measures are:  $u = \frac{m}{m_1}$ ,  $v = \frac{m_2}{m_1}$ ,  $w = \frac{m_2}{m}$ .

## 17.3 Normative Measures of Inequality

Normative measures evaluate the loss of *Social Welfare* due to inequality. Assume a Social Welfare Function (SWF) where  $W = \sum U(y_i)$ , and  $U(y_i)$  is strictly increasing and concave.

### Definition 9: Equally Distributed Equivalent Income ( $y_{ede}$ )

The specific income that, if given equally to everyone, generates the exact same total Social Welfare as the current unequal distribution.

$$W(y_{ede}, y_{ede}, \dots, y_{ede}) = W(y_1, y_2, \dots, y_n) \quad (63)$$

*Note: Due to concave utility,  $y_{ede} < m$  (the arithmetic mean). Maximum welfare requires  $y_{ede} = m$ .*

### Dalton vs. Atkinson Measures

**Dalton's Measure ( $I_D$ ):**  $I_D = 1 - \frac{\sum U(y_i)}{nU(m)} = 1 - \frac{U(y_{ede})}{U(m)}$

*Drawback:* Variant under linear transformations of utility.

**Atkinson's Measure ( $I_A$ ):**  $I_A = 1 - \frac{y_{ede}}{m}$

*Advantage:* Invariant under any positive linear transformation of the utility function.

### Theorem 17: Atkinson's Theorem

For two communities with equal population and mean income: If the Lorenz Curve of Community A lies entirely inside (above) the Lorenz Curve of Community B, Community A has unambiguously higher social welfare, regardless of the specific utility function. If the curves cross, ranking requires a specific utility function.

## 17.4 Macroeconomic Theory & The Indian Context

**1. Kuznets' Inverted-U Hypothesis:** As a country's per capita income increases, inequality initially **ris**es, reaches a peak, and eventually **falls**, forming an inverted U-shaped curve.

### 2. Exam Points: Causes of Inequality in India

- **Unemployment:** Inadequate job creation in planned development.
- **Inflation:** Real wages in the unorganized sector fall while profit-earners gain.
- **Tax Evasion:** High direct tax rates led to black money concentration.
- **Regressive Taxation:** Over-reliance on indirect taxes (sales/excise) which disproportionately burden the poor.
- **Green Revolution:** Benefited only rich landowners who could afford modern agricultural inputs.

### 3. Exam Points: Measures Taken to Reduce Inequality

- **Land Reforms:** Abolition of Zamindari and imposition of land ceilings.
- **Monopoly Control:** MRTP Act (1969) / Competition Act (2009) to restrict trade monopolies.
- **Social Security:** Employees State Insurance, Provident Funds, and Maternity Benefits (mostly restricted to urban/organized sectors).
- **Employment Schemes:** Poverty alleviation through IRDP, NREP, and MGNREGS.
- **Fiscal Policy:** Progressive direct taxation and transfer payments (pensions, subsidies).

## 18 Normative Measures of Inequality

### 18.1 Social Welfare Function (SWF) & Utility

Normative measures evaluate inequality based on the loss of social welfare. Let  $x_1, x_2, \dots, x_n$  be individual incomes.

- **Individual Utility:**  $U(x_i)$  is assumed to be strictly increasing ( $U' > 0$ ) and strictly concave ( $U'' < 0$ , reflecting diminishing marginal utility of money).
- **Social Welfare Function (SWF):** An additively separable and symmetric function of individual utilities:  $W = \sum_{i=1}^n U(x_i)$ .

#### Theorem 18: Jensen's Inequality (Applied to SWF)

If  $U(x)$  is a concave function, the weighted mean of utilities is less than or equal to the utility of the weighted mean. Assuming equal weights ( $1/n$ ):

$$\frac{\sum U(x_i)}{n} \leq U\left(\frac{\sum x_i}{n}\right) \implies \sum U(x_i) \leq nU(\bar{x}) \quad (64)$$

*Implication:* Social Welfare attains its absolute maximum when total income is distributed perfectly equally.

### 18.2 Dalton's and Atkinson's Measures

#### Definition 10: Dalton's Measure ( $I_D$ )

Measures the gap between actual welfare and maximum potential welfare (perfect equality).

$$I_D = 1 - \frac{\sum U(x_i)}{nU(\bar{x})} \quad (65)$$

*Critical Flaw:* It is not invariant under positive linear transformations of the utility function (i.e., changing  $U$  to  $a + bU$  changes the index value).

#### Definition 11: Atkinson's Measure ( $I_A$ ) & $x_{EDE}$

Atkinson resolved Dalton's flaw by using **Equally Distributed Equivalent Income** ( $x_{EDE}$ ), defined as the per-capita income that yields the exact same total welfare as the current unequal distribution:  $nU(x_{EDE}) = \sum U(x_i)$ .

$$I_A = 1 - \frac{x_{EDE}}{\bar{x}} \quad (66)$$

*Advantage:*  $I_A$  is perfectly invariant under positive linear transformations of  $U(x)$ .

### 18.3 The Utility Function and Inequality Aversion (epsilon)

For  $I_A$  to be invariant to proportional shifts in income (Scale Independence Axiom), the utility function must take a highly specific mathematical form:

$$U(x) = \begin{cases} A + B \frac{x^{1-\epsilon}}{1-\epsilon} & \text{for } \epsilon > 0, \epsilon \neq 1 \\ A + B \ln(x) & \text{for } \epsilon = 1 \end{cases} \quad (67)$$

Here,  $\epsilon$  is the parameter of inequality aversion. Higher  $\epsilon$  means society is more sensitive to transfers among the poor.

**Atkinson's Final Explicit Formula:**

$$I_A = 1 - \left[ \frac{1}{n} \sum_{i=1}^n \left( \frac{x_i}{\bar{x}} \right)^{1-\epsilon} \right]^{\frac{1}{1-\epsilon}} \quad (68)$$

#### Exam Point: Specific Values of $\epsilon$

How  $I_A$  changes based on the inequality aversion parameter:

- **If  $\epsilon = 0$ :**  $I_A = 0$  (The measure is completely insensitive to distribution).
- **If  $\epsilon = 1$ :**  $I_A = 1 - \frac{GM}{AM}$  (where GM = Geometric Mean).
- **If  $\epsilon = 2$ :**  $I_A = 1 - \frac{HM}{AM}$  (where HM = Harmonic Mean).
- **If  $\epsilon \rightarrow \infty$ :**  $I_A \rightarrow 1 - \frac{\min(x_1, \dots, x_n)}{AM}$  (Determined entirely by the poorest person).

*Mathematical Fact:* Because  $AM > GM > HM > \text{Min}$ ,  $I_A$  strictly increases as  $\epsilon$  increases.

## 18.4 Lorenz Curve Comparisons

### Theorem 19: Atkinson's Theorem (Hardy-Littlewood-Polya)

For two distributions with the same mean: If the Lorenz Curve (LC) of Distribution I lies uniformly above (or on) the LC of Distribution II without crossing, then  $W_1 > W_2$  for **every** strictly concave utility function.

*Rule:* Non-intersecting Lorenz Curves provide unambiguous inequality rankings. If the curves intersect, rankings depend on the specific utility function chosen.

## 18.5 Exam Point: Axiomatic Check of Conventional Measures

How standard measures hold up against the strict axioms of inequality:

- **Relative Mean Deviation (RMD): Fails.** It is completely insensitive to income transfers that happen on the same side of the mean.
- **Coefficient of Variation (CV):** Satisfies Principle of Transfer, but **Fails** Principle of Diminishing Transfer. (The reduction in inequality  $\Delta I$  is the same regardless of whether the transfer is between rich or poor).
- **Lorenz Ratio (LR / Gini):** Satisfies Principle of Transfer, but **Fails** Principle of Diminishing Transfer. ( $\Delta I$  is actually greater for transfers affecting the middle class, not the extreme poor).
- **Standard Deviation of Logs ( $SD(Ln)$ ):** Satisfies both Principle of Transfer and Diminishing Transfer.
- **Atkinson's Measure ( $I_A$ ):** Satisfies both Principle of Transfer and Diminishing Transfer (for  $\epsilon > 0$ ).

## 19 Measures of Poverty

### 19.1 Fundamental Concepts

#### Absolute vs. Relative Poverty

**Absolute Poverty:** A family is poor if its total earning is insufficient to obtain minimum biological/nutritional necessities (basic needs like food, shelter, clothing) for survival.

**Relative Poverty:** Based on relative deprivation. A family is poor if it cannot share in the standard diets, customs, and activities of the society they live in. *Note: Proportional increases in everyone's income will not change relative poverty, but will drastically reduce absolute poverty.*

#### Methods of Identification:

- **Direct Method:** Identifies households whose actual consumption basket leaves basic nutritional needs unsatisfied (revealed via budget data).
- **Income Method:** Specifies a threshold income ( $Z$ , the poverty line). Anyone with income  $< Z$  is poor. *Advantage:* It treats people who have the *ability* to meet their needs (but choose not to) as non-poor.

**Household Composition Adjustment** Because household size and composition (age/sex) vary, we use the **Adult Equivalent Scale**. One adult male is the standard unit, and other members are expressed as fractions of an equivalent adult. Family income is divided by this number, not just the head count.

### 19.2 Basic Measures of Poverty

#### Notation Setup:

- $n$  = Total population.
- $q$  = Number of poor persons.
- $Z$  = Poverty line income.
- $y_i$  = Income of the  $i^{th}$  poor person (sorted:  $y_1 \geq y_2 \cdots \geq y_q$ ).
- $g_i = Z - y_i$  = Poverty gap of the  $i^{th}$  poor person.
- $\bar{y}_q = \frac{1}{q} \sum y_i$  = Average income of the poor.

#### 1. Head Count Ratio (H):

$$H = \frac{q}{n} \quad (69)$$

*Flaw:* Measures *how many* are poor, but ignores *how much* they suffer (the poverty gap).

#### 2. Income Gap Ratio (I):

$$I = \frac{1}{q} \sum_{i=1}^q \frac{(Z - y_i)}{Z} = 1 - \frac{\bar{y}_q}{Z} \quad (70)$$

*Flaw:* Measures the average depth of poverty, but ignores *how many* are poor.

**The Joint Flaw of H and I:** Both measures completely ignore the **inequality of incomes** among the poor.

### 19.3 Sen's Index of Poverty

Sen proposed that a proper index must satisfy two core axioms:

#### Theorem 20: Sen's Core Axioms

1. **Monotonicity Axiom (M):** An increase in the income of any poor person must strictly reduce the poverty measure.
2. **Weak Transfer Axiom (WT):** A transfer of income from a poorer person to a richer person (who remains below the poverty line) must strictly increase the poverty measure.

To build his index, Sen used two additional mathematical axioms:

- **Axiom R (Ranked Relative Deprivation):** The weight given to a person's poverty gap is their income rank among the poor ( $v_i = i$ ).
- **Axiom A (Normalized Absolute Deprivation):** If all poor people have the exact same income, the index must equal  $H \times I$ .

#### Definition 12: Sen's Poverty Index (P)

Combining the axioms, Sen's index incorporates  $H$ ,  $I$ , and  $G$  (the Gini coefficient of the incomes of the poor):

$$P = H[I + (1 - I)G] \quad (71)$$

*Alternative Form:*  $P = H \left[ 1 - \frac{\bar{y}_q(1-G)}{Z} \right]$ .

Sen's Index satisfies both the M and WT axioms.

### 19.4 Other Poverty Indices

- **Sudhir Anand's Index ( $P_{SA}$ ):** Modifies the Income Gap Ratio by expressing it as a fraction of the *community's mean income* ( $\mu$ ), not  $Z$ .

$$P_{SA} = \frac{H}{\mu} [Z - \bar{y}_q(1 - G)] \quad (72)$$

- **Kakwani's Index ( $P_K$ ):** Weight is proportional to the income of all poor persons richer than person  $i$ .

$$P_K = \frac{H}{\mu} \left[ Z - \frac{\bar{y}_q}{1 + G} \right] \quad (73)$$

- **Blackorby & Donaldson Index ( $P_{BD}$ ):** Uses  $e$ , the equally distributed equivalent income of the poor.

$$P_{BD} = H \left( 1 - \frac{e}{Z} \right) \quad (74)$$

- **Thon's Index ( $P_T$ ):** Ranks the poor person against the *entire community*, ensuring the index increases for all transfers from a poorer to a richer person.

### 19.5 Decomposition of Poverty Measures

To target anti-poverty programs, an index should be additively decomposable across  $k$  mutually exclusive subgroups (where  $n_i$  is the population of subgroup  $i$ ):

$$P = \sum_{i=1}^k \frac{n_i}{n} P_i \quad (75)$$

Note: Sen's Index is NOT additively decomposable. The Head Count Ratio (H) IS decomposable.

**1. Chakravarty's Index ( $P_{SRC}$ ):**

$$P_{SRC} = \frac{1}{n} \sum_{i=1}^q \left[ 1 - \left( \frac{y_i}{Z} \right)^e \right], \quad 0 < e < 1 \quad (76)$$

**2. Foster, Greer, and Thorbecke (FGT) Index:**

$$P_{FGT} = \frac{1}{n} \sum_{i=1}^q \left( \frac{Z - y_i}{Z} \right)^\alpha, \quad \alpha \geq 0 \quad (77)$$

Properties of FGT:

- If  $\alpha = 0$ ,  $P_{FGT} = H$  (Head Count Ratio).
- If  $\alpha = 1$ ,  $P_{FGT} = HI$  (Poverty Gap Index).
- If  $\alpha = 2$ , it is the **Squared Poverty Gap Index** (highly sensitive to the poorest of the poor).

## 19.6 Exam Points: Poverty Estimation in India

**The Tendulkar Expert Group Report (2009) Salient Features:**

- **Uniform PLB:** Adopted a uniform Poverty Line Basket (PLB) for both rural and urban sectors based on the 61<sup>st</sup> round NSS urban data.
- **Shift from Calorie Norms:** Moved away from strict calorie norms, standardizing around 1776 Kcal/day (close to FAO's 1800 Kcal).
- **MRP over URP:** Adopted the Mixed Reference Period (MRP) instead of the Uniform Reference Period (URP) for estimates.
- **Internal Price Adjustments:** Used the **Fisher Index** for updating poverty lines to properly weight health and education.
- **Base Year:** Reset the base year to 2004-05.

## 20 Demand Analysis: Core Concepts

### 20.1 The Optimum Quantities & Demand Function

A consumer maximizes their preference (utility) function  $u(q_1, q_2, \dots, q_n)$  subject to a fixed disposable income  $x$  and fixed prices  $p_1, p_2, \dots, p_n$ .

**Optimization via Lagrange Multiplier:** We maximize the Lagrangian:  $L = u + \lambda(x - \sum_{i=1}^n p_i q_i)$ . Setting partial derivatives to zero yields:

$$\frac{\partial L}{\partial q_i} = u_i - \lambda p_i = 0 \quad \forall i \implies \frac{u_1}{p_1} = \frac{u_2}{p_2} = \dots = \frac{u_n}{p_n} = \lambda \quad (78)$$

Solving this system of equations yields the **Marshallian Demand Function** for the  $i^{th}$  good:

$$q_i = f_i(p_1, p_2, \dots, p_n, x) \quad (79)$$

## 20.2 Demand Curve vs. Engel Curve

Special cases of the demand function are formed by holding certain parameters constant (partial equilibrium analysis).

### Exam Point: Types of Curves

**1. Demand Curve:** Indicates the *ceteris paribus* effect of a change in a good's **own price** on its quantity demanded.

- **Held Constant:** Income ( $x$ ) and all other prices ( $p_j, j \neq i$ ).
- **Equation:**  $q_i = f_i(p_i)$

**2. Engel Curve (Income-Consumption Relation):** Indicates the effect of a change in **income** on the quantity/expenditure of the  $i^{th}$  good.

- **Held Constant:** All prices ( $p_1, \dots, p_n$ ).
- **Equation:**  $q_i = g_i^{**}(x)$  or  $p_i q_i = g_i^*(x)$

## 20.3 Elasticities of Demand

Elasticities measure responsiveness as percentage changes, making them independent of measurement units.

**1. Own Price Elasticity ( $\eta_{jj}$ ):** Percentage change in quantity demanded for a 1% change in its own price.

$$\eta_{jj} = \frac{\partial f_j}{\partial p_j} \cdot \frac{p_j}{q_j} = \frac{\partial \ln(f_j)}{\partial \ln(p_j)} \quad (80)$$

*Classification:* Price elastic if  $|\eta_{jj}| > 1$ , price inelastic if  $|\eta_{jj}| < 1$ .

**2. Income Elasticity ( $\eta_{jx}$ ):** Percentage change in quantity demanded for a 1% change in income.

$$\eta_{jx} = \frac{\partial f_j}{\partial x} \cdot \frac{x}{q_j} = \frac{\partial \ln(f_j)}{\partial \ln(x)} \quad (81)$$

*Classification:* Income elastic if  $\eta_{jx} > 1$ , income inelastic if  $\eta_{jx} < 1$ .

**3. Cross-Price Elasticity ( $\eta_{jj'}$ ):** Effect of a change in the price of the  $j^{th}$  good on the demand for the  $j^{th}$  good.

$$\eta_{jj'} = \frac{\partial f_j}{\partial p_{j'}} \cdot \frac{p_{j'}}{q_j} = \frac{\partial \ln(f_j)}{\partial \ln(p_{j'})} \quad (82)$$

## 20.4 Classification of Goods & Engel's Law

- **Engel's Law:** The proportion of income spent on food (or any *necessary/income inelastic* good) strictly decreases as total income increases.
- **Normal vs. Giffen Good:**
  - *Normal:* Downward sloping demand curve (Negative own price elasticity).
  - *Giffen:* Upward sloping demand curve (Positive own price elasticity).
- **Superior vs. Inferior Good:**
  - *Superior:* Upward sloping Engel curve (Positive income elasticity).
  - *Inferior:* Downward sloping Engel curve (Negative income elasticity).

*Rule:* A Giffen good **cannot** be a superior good.

## 20.5 Mathematical Restrictions on Demand Functions

Economic theory dictates that demand functions must satisfy four strict mathematical conditions.

*Notation:* Let  $s_i = \frac{p_i q_i}{x}$  be the **budget share** (proportion of income spent on good  $i$ ).

### Theorem 21: 1. Budget Constraint

The total expenditure across all goods must equal total income. Dividing by  $x$  yields:

$$\sum_{i=1}^n s_i = 1 \quad (83)$$

### Theorem 22: 2. Adding Up Criterion (Engel Aggregation)

Found by differentiating the budget constraint with respect to income ( $x$ ). The weighted mean of income elasticities (where weights are budget shares) must add up to 1.

$$\sum_{i=1}^n s_i \eta_{ix} = 1 \quad (84)$$

### Theorem 23: 3. Cournot Aggregation Conditions

Found by differentiating the budget constraint with respect to price  $p_i$ . There are  $n$  such conditions (one for each price).

$$\sum_{j=1}^n \eta_{ji} \frac{s_j}{s_i} + 1 = 0 \quad (85)$$

### Theorem 24: 4. Homogeneity Condition (Euler's Theorem)

A proportional scaling of all prices and income by factor  $\alpha$  does not change demand. Thus, the demand function is **homogeneous of degree zero**. Applying Euler's Theorem gives:

$$\sum_{j=1}^n \eta_{ij} + \eta_{ix} = 0 \quad (86)$$

*Meaning:* The sum of all own and cross-price elasticities for a good, plus its income elasticity, exactly equals zero.

## 21 Empirical Demand Analysis

### 21.1 Types of Data in Demand Analysis

- **Cross-Section (CS) Data:** Data collected at a single, fixed point in time from multiple individual households.
- **Time Series (TS) Data:** Data collected from a single entity (or aggregate) across various points in time.
- **Panel Data (Continuous CS):** Combines CS and TS by observing the *exact same* individuals/households at multiple points in time.

## 21.2 The Engel Curve & Classification of Goods

Assuming prices are constant across a small region, consumption  $y_i$  depends only on total expenditure/income  $x$  and household preferences. If there is **no economy of scale** within the household, we use the **per capita formulation**:  $y = g(x)$ .

### Theorem 25: Classification of Goods by Engel Elasticity ( $\eta_{ix}$ )

Let  $\eta_{ix}$  be the income elasticity of demand.

1. **Inferior Goods:**  $\frac{\partial y_i}{\partial x} < 0$ . As income increases, consumption decreases.
2. **Necessary Goods:**  $0 < \eta_{ix} < 1$ . As income increases, consumption increases, but at a decreasing rate ( $\frac{\partial^2 y_i}{\partial x^2} < 0$ ).
3. **Luxury Goods:**  $\eta_{ix} > 1$ . As income increases, consumption increases at an accelerating rate ( $\frac{\partial^2 y_i}{\partial x^2} > 0$ ).

## 21.3 Quantity vs. Quality Elasticities

If consumption is measured in value terms ( $v = q \times p$ ), where  $q$  is quantity and  $p$  is price:

$$\eta_v = \eta_q + \eta_p \quad (87)$$

Where:

- $\eta_v$  = Value elasticity of income.
- $\eta_q$  = Quantity elasticity of income.
- $\eta_p$  = Quality elasticity (Rate at which consumers shift to finer/more expensive qualities as income rises).

*Adding Up Criterion (Value Terms):* The weighted average of Engel elasticities for all items equals 1 ( $\sum w_i \eta_{ix} = 1$ ).

## 21.4 Criteria for Choosing an Engel Curve

### 1. Economic Criteria:

- **Threshold Level ( $x_0$ ):** The income level at which consumption begins (primarily for luxury goods).
- **Satiety Level ( $y_0$ ):** The absolute upper bound of physical consumption (primarily for necessary goods).
- **Sigmoid Form:** The ideal Engel curve follows an S-shape (Luxury  $\rightarrow$  Necessary  $\rightarrow$  Inferior as income spans 0 to  $\infty$ ).

**2. Statistical Criteria (Model Selection):** To avoid arbitrarily increasing the  $R^2$  value by adding parameters, we penalize complexity using Information Criteria:

### Exam Point: AIC and BIC Formulas

Let  $k$  = number of parameters,  $n$  = number of observations, and  $\hat{L}$  = maximized likelihood.

- **Akaike Information Criterion (AIC):**  $AIC = 2k - 2 \ln(\hat{L})$
- **Bayesian Information Criterion (BIC):**  $BIC = k \ln(n) - 2 \ln(\hat{L})$

*Rule:* The preferred model is always the one with the **minimum** AIC or BIC value (Parsimony).

## 21.5 Popular Forms of Engel Curves

### A. Standard Two-Parameter Forms:

- **Linear:**  $y = \alpha + \beta x$  ( $\beta$  = *Marginal Propensity to Consume*).
- **Double Log:**  $\ln(y) = \alpha + \beta \ln(x)$  ( $\beta$  = *Income Elasticity*. *Constant elasticity; fits luxury goods well but lacks threshold/satiety*).

### B. Tornquist's Forms (For Classification):

- **Necessary Goods:**  $y = \frac{\alpha x}{x+\beta}$  ( $\beta > 0$ ).  
*Properties:* No threshold. Satiety level =  $\alpha$ .
- **Luxury Goods:**  $y = \frac{\alpha(x-\gamma)}{x+\beta}$  ( $x > \gamma, \beta > 0$ ).  
*Properties:* Threshold level =  $\gamma$ . Satiety level =  $\alpha$ .
- **Inferior Goods:**  $y = \frac{\alpha x}{x-\beta}$  ( $x > \beta, \beta > 0$ ).  
*Properties:* As  $x \rightarrow \infty, y \rightarrow \alpha$ .

## 21.6 Specific Concentration Curve (SCC) & Iyengar's Method

The SCC ( $Q_y(x)$ ) relates the expected consumption of a specific commodity to the cumulative population distribution.

### Theorem 26: Advantages of the SCC Approach over Least Squares (LS)

1. **Consistency:** In grouped data, LS estimates of log-linear models are mathematically inconsistent due to averaging. Iyengar's SCC method yields consistent estimates.
2. **Zero-Value Problem:** LS cannot handle  $y = 0$  observations because  $\ln(0) = -\infty$ . SCC easily incorporates zero-expenditure households without discarding data.

## 21.7 Demand Projection from Cross-Section Data

If the Engel function  $E(y|x) = ax^b$  remains constant, and we assume everybody's income increases by proportion  $c$  (i.e.,  $x \rightarrow x(1+c)$ ) and the population grows at rate  $\alpha$ :

$$\frac{\text{Aggregate Target Demand}}{\text{Aggregate Base Demand}} = \frac{P^* E^*(y)}{PE(y)} = (1+c)^b (1+\alpha) \quad (88)$$

### Exam Point: Lognormal Relaxation

If we assume income follows a Lognormal distribution, mean income increases by  $c$ , and overall inequality (Lorenz Ratio) strictly **decreases** ( $\sigma^{*2} < \sigma^2$ ), the direction of per capita consumption  $E^*(y)/E(y)$  depends entirely on the elasticity  $b$ :

- **Case 1** ( $0 < b < 1$ ): Necessary goods. Ratio  $> 1$  (Consumption strictly increases).
- **Case 2** ( $b < 0$ ): Inferior goods. Ratio  $< 1$  (Consumption strictly decreases).
- **Case 3** ( $b > 1$ ): Luxury goods. Direction is mathematically ambiguous (can be  $< 1$ ,  $= 1$ , or  $> 1$ ).

## 22 Empirical Demand Analysis: System of Equations

### 22.1 System of Demand Equations

A complete system of demand equations analyzes the interdependence among  $n$  commodities in a market, particularly cross-price effects.

$$y_j = f_j(p_1, p_2, \dots, p_n, x, u_j) \quad \text{for } j = 1, 2, \dots, n \quad (89)$$

*Variables:*  $y_j$  (quantities consumed) are endogenous. Prices  $(p_1, \dots, p_n)$  and total income  $(x)$  are exogenous.

**Common Functional Forms:**

- **Linear System:**  $y_j = a_j + \sum_k b_{jk} p_k + c_j x + e_j$
- **Log-Linear (Constant Elasticity):**  $\ln(y_j) = a_j + \sum_k b_{jk} \ln(p_k) + c_j \ln(x) + e_j$

### 22.2 Linear Expenditure System (LES)

#### Definition 13: Linear Expenditure System (LES)

The LES decomposes expenditure on good  $j$  into two components: committed (subsistence) expenditure and discretionary expenditure.

$$p_j y_j = p_j y_j^0 + \beta_j \left( x - \sum_k p_k y_k^0 \right) + e_j \quad (90)$$

Where:

- $y_j - y_j^0 > 0$ ,  $0 < \beta_j < 1$ ,  $\sum \beta_j = 1$
- $p_j y_j^0$ : **Committed Expenditure** (minimum needed for base amount).
- $(x - \sum p_k y_k^0)$ : **Supernumerary Income** (income left after buying base amounts of all goods).
- $\beta_j$ : **Marginal Budget Share** (fraction of supernumerary income spent on good  $j$ ).

### Exam Point: Properties of LES

1. **Theoretical Compliance:** It automatically satisfies **all** theoretical restrictions: Budget constraint, Homogeneity, Adding-Up Criterion (AUC), Cournot aggregation, Slutsky negativity, and Slutsky symmetry.
2. **Utility Origin:** It can be derived directly from maximizing the **Stone-Geary (Klein-Rubin) Utility Function:**  $U = \sum \beta_j \ln(y_j - y_j^0)$
3. **Shape:** Dividing the LES by  $p_j$  yields the demand equation, which is **hyperbolic** in own price and **linear** in income.

**Estimation Challenges & Approaches:** Estimation is non-linear in parameters ( $\beta_j \times y_k^0$ ). Multicollinearity among prices is a major issue. Four standard estimation approaches:

1. Fix base quantities ( $y_k^0$ ) extraneously, then estimate  $\beta_j$  via simple linear regression.
2. Fix marginal shares ( $\beta_j$ ) extraneously, then estimate base quantities  $y_k^0$ .
3. Iterative approach: Alternate between fixing  $y_k^0$  and  $\beta_j$  to minimize Total Sum of Squares (TSS).
4. Simultaneous Grid Search: Test all possible values to minimize TSS globally.

### 22.3 Effect of Household Size and Scale Economies

Instead of simple per-capita formulation ( $\frac{y}{n} = f(\frac{x}{n})$ ), we use a generalized form incorporating household size  $n$ :

$$y = ax^bn^\gamma \implies \frac{y}{n} = a\left(\frac{x}{n}\right)^b n^\delta \quad (91)$$

Where  $\delta = \gamma + b - 1$ .

#### Theorem 27: Conditions for Economy of Scale

By estimating  $\ln(y) = \ln(a) + \gamma \ln(n) + b \ln(x)$ , we test the value of  $\delta$  to determine scale effects (e.g., bulk purchasing discounts or shared utilities like electricity):

- $\gamma + b < 1 \implies \delta < 0 \implies$  **Economy of Scale** (Consumption increases less than proportionately as  $n$  increases).
- $\gamma + b > 1 \implies \delta > 0 \implies$  **Diseconomy of Scale**.
- $\gamma + b = 1 \implies \delta = 0 \implies$  **Independent of Scale**.

### 22.4 Prais-Houthakker (PH) Formulation

PH refines scale economies by assuming a household of size  $n$  has an **effective size** of  $n^{\theta_i}$  for specific commodity  $i$ , and  $n^{\theta_0}$  for total consumption.

$$\frac{y_i}{n^{\theta_i}} = f\left(\frac{x}{n^{\theta_0}}\right) \quad (92)$$

The ratio  $\frac{\theta_i}{\theta_0}$  reflects the economy/diseconomy of scale for commodity  $i$  relative to total consumption.

### Theorem 28: PH Parameter Restrictions

Assuming an increase in  $n$  (while holding total consumption  $x$  fixed) only redistributes consumption ( $\sum \frac{\partial y_i}{\partial n} = 0$ ), and an increase in  $x$  is fully distributed ( $\sum \frac{\partial y_i}{\partial x} = 1$ ):

$$\sum w_i \eta_{ix} = 1 \quad \text{and} \quad \sum w_i \eta_{in} = 0 \quad (93)$$

Because  $\eta_{in} = \theta_i - \eta_{ix}\theta_0$ , substituting this into the second restriction yields the core identity relating the scale parameters:

$$\theta_0 = \sum w_i \theta_i \quad \text{or} \quad 1 - \theta_0 = \sum w_i (1 - \theta_i) \quad (94)$$

**PH Method of Estimation (The Two-Step Quality Index Method):** Because  $y_i = a_i x^{b_i} n^{\theta_i - b_i \theta_0}$  cannot separate  $\theta_i$  from  $\theta_0$ , PH uses a two-step workaround:

1. **Step 1 (Find  $\theta_0$ ):** Define a quality index  $P_n^j$  (the ratio of prices paid by household  $j$  compared to an average household). Regress  $P_n$  on  $\ln(x)$  and  $\ln(n)$ :

$$P_n = \alpha + \beta \ln(x) - \beta \theta_0 \ln(n) \quad (95)$$

Estimate  $\hat{\beta}$  and the coefficient on  $\ln(n)$  to isolate and solve for  $\hat{\theta}_0$ .

2. **Step 2 (Find  $\theta_i$ ):** Run a single equation regression for commodity  $i$ :

$$\ln(y_i) = \ln(a_i) + b_i \ln(x) + c_i \ln(n) \quad (96)$$

Since  $\hat{c}_i = \hat{\theta}_i - \hat{b}_i \hat{\theta}_0$ , and we already know  $\hat{\theta}_0$  and  $\hat{b}_i$ , we can algebraically solve for the specific scale parameter  $\hat{\theta}_i$ .

## 23 The Production Function

### 23.1 Basic Concept and Marginal Products

A **Production Function** represents the technical relationship between the quantities of inputs and the maximum output that can be produced. Assuming two factors of production, Capital ( $K$ ) and Labour ( $L$ ):

$$Q = f(K, L) \quad (97)$$

### Theorem 29: Classical Assumptions (Diminishing Marginal Returns)

The classical theory assumes that Marginal Products (MP) are strictly positive, but they diminish as more of a single input is added. Mathematically, for a twice-differentiable function:

- **Positive Marginal Products:**  $Q_K = \frac{\partial Q}{\partial K} > 0$  and  $Q_L = \frac{\partial Q}{\partial L} > 0$
- **Diminishing Marginal Products:**  $Q_{KK} = \frac{\partial^2 Q}{\partial K^2} < 0$  and  $Q_{LL} = \frac{\partial^2 Q}{\partial L^2} < 0$

### 23.2 Marginal Rate of Substitution (MRS)

An **isoquant** represents all combinations of  $K$  and  $L$  that produce a constant level of output ( $f(K, L) = \text{Constant}$ ). Taking the total differential along an isoquant:

$$Q_K dK + Q_L dL = 0 \implies -\frac{dK}{dL} = \frac{Q_L}{Q_K} = R > 0 \quad (98)$$

*Key Properties:*

- $R$  is the **Marginal Rate of Substitution**. It is the absolute value of the slope of the isoquant.
- Isoquants are **convex to the origin**, meaning  $dK/dL$  is decreasing, making the second derivative strictly positive:  $\frac{d^2K}{dL^2} > 0$ .

### 23.3 Elasticity of Substitution (sigma)

The elasticity of substitution measures the ease with which capital can be substituted for labour. It is defined as the proportionate change in the input ratio ( $K/L$ ) divided by the proportionate change in the MRS ( $R$ ).

#### Definition 14: Elasticity of Substitution ( $\sigma$ )

$$\sigma = \frac{d(K/L)/(K/L)}{dR/R} = \frac{d \ln(K/L)}{d \ln(R)} \quad (99)$$

*Alternative Expressions (Exam Essentials):*

$$\sigma = \frac{R(LR + K)}{KL \left( \frac{d^2R}{dL^2} \right)} \quad (100)$$

$$\sigma = \frac{Q_K Q_L (K Q_K + L Q_L)}{-KL (Q_K^2 Q_{LL} + Q_L^2 Q_{KK} - 2Q_K Q_L Q_{KL})} \quad (101)$$

*Limits:* Since  $R, L, K$ , and the curvature are all positive,  $\sigma \geq 0$ .

- If  $\sigma = 0$ : Substitution is impossible (Leontief/L-shaped isoquants).
- If  $\sigma = \infty$ : Perfect substitution (Linear isoquants).

### 23.4 Constant Returns to Scale & Euler's Theorem

A production function exhibits **Constant Returns to Scale** if increasing all inputs by a factor  $\lambda$  increases output by the exact same factor  $\lambda$ :

$$f(\lambda K, \lambda L) = \lambda f(K, L) = \lambda Q \quad (102)$$

This means the function is **homogeneous of degree one**.

#### Theorem 30: Euler's Theorem Application

If  $Q$  is homogeneous of degree 1, applying Euler's Theorem gives:

$$Q = K Q_K + L Q_L \quad (103)$$

*Economic Meaning:* If input factors are paid exactly their marginal products, the total product  $Q$  is perfectly exhausted between them. *Mathematical identities resulting from this:*  $K Q_{KK} + L Q_{LK} = 0$  and  $L Q_{LL} + K Q_{KL} = 0$

## 24 Economic Models of the Firm

Firms operate under specific market constraints. Let  $r$  = price of capital,  $w$  = price of labour, and  $p$  = price of output.

### 24.1 Model 1: Cost Minimization

The firm minimizes total cost  $C = rK + wL$  subject to an exogenously given output  $Q_0 = f(K, L)$ .

- **Lagrangian:**  $L = rK + wL - \lambda(Q_0 - f(K, L))$
- **First Order Conditions (FOC):**  $\frac{\partial L}{\partial K} = r - \lambda f_K = 0$  and  $\frac{\partial L}{\partial L} = w - \lambda f_L = 0$
- **Optimum Solution:**

$$\frac{f_L}{f_K} = \frac{w}{r} \quad (104)$$

*Conclusion:* Cost is minimized when the Marginal Rate of Substitution equals the factor price ratio.

### 24.2 Model 2: Profit Maximization (Perfect Competition)

The firm maximizes profit  $\pi = pQ - rK - wL$  subject to  $Q = f(K, L)$ . Because the market is perfectly competitive, the firm is a price-taker ( $p, r, w$  are constants).

- **FOC (Marginal Productivity Conditions):**

$$r = pf_K \quad \text{and} \quad w = pf_L \quad (105)$$

*Conclusion:* The value of the marginal product of each input must equal its price.

- **Second Order Conditions (SOC):** For  $\pi$  to be a true maximum, the principal minors of the Hessian matrix must alternate in sign:

$$Q_{KK} < 0, \quad Q_{LL} < 0, \quad \text{and} \quad (Q_{KK}Q_{LL} - Q_{KL}^2) > 0 \quad (106)$$

### 24.3 Model 3: Profit Maximization (Imperfect Competition)

If the firm faces a downward-sloping demand curve, price  $p$  is no longer constant. Assume a constant elasticity demand curve:  $p = bQ^\eta$  (where  $1/\eta$  is the elasticity of demand). Total Revenue (TR) is  $pQ = bQ^{\eta+1}$ .

- **Lagrangian:**  $\pi = bQ^{\eta+1} - rK - wL - \lambda(Q - f(K, L))$
- **FOC:**

$$w = \lambda f_L, \quad r = \lambda f_K \implies \frac{w}{r} = \frac{f_L}{f_K} \quad (107)$$

$$\lambda = (\eta + 1)bQ^\eta \quad (\text{when } Q \text{ is a variable}) \quad (108)$$

## 25 Appendix: General Optimization Conditions

### Exam Point: Bordered Hessian Rules

For a general Lagrangian optimization  $F = f + \lambda g$ , we look at the Bordered Hessian determinants  $|H_k|$ .

- **For Maximization:** The determinants must alternate in sign, starting positive.

$$|H_2| > 0, \quad |H_3| < 0, \quad |H_4| > 0 \dots$$

- **For Minimization:** All determinants must be strictly negative.

$$|H_2| < 0, \quad |H_3| < 0, \quad |H_4| < 0 \dots$$

*Application to Profit Max SOC:*

$$\begin{vmatrix} \pi_{LL} & \pi_{LK} & f_L \\ \pi_{KL} & \pi_{KK} & f_K \\ f_L & f_K & 0 \end{vmatrix} > 0 \quad (109)$$

Which expands to:  $D = 2f_{KL}f_Lf_K - f_{LL}f_K^2 - f_{KK}f_L^2 > 0$ .

## 26 The Cobb-Douglas Production Function

### 26.1 Introduction and Core Properties

The Cobb-Douglas (CD) production function is the most widely used mathematical form in production analysis. For two inputs, Capital ( $K$ ) and Labour ( $L$ ), it is defined as:

$$Q = AK^\alpha L^\beta \quad (110)$$

Where:  $A$  represents the efficiency of the production process, and  $\alpha, \beta$  are the output elasticities of capital and labour, respectively.

### Theorem 31: Core Properties of the Cobb-Douglas Function

#### Property 1: Positive and Diminishing Marginal Products

- $Q_K = \frac{\alpha Q}{K} > 0$  and  $Q_L = \frac{\beta Q}{L} > 0 \implies \alpha > 0, \beta > 0$
- $Q_{KK} = \alpha(\alpha - 1) \frac{Q}{K^2} < 0 \implies 0 < \alpha < 1$
- $Q_{LL} = \beta(\beta - 1) \frac{Q}{L^2} < 0 \implies 0 < \beta < 1$

#### Property 2: Returns to Scale The function is homogeneous of degree $n = \alpha + \beta$ .

- $\alpha + \beta = 1 \implies$  Constant returns to scale.
- $\alpha + \beta < 1 \implies$  Decreasing returns to scale.
- $\alpha + \beta > 1 \implies$  Increasing returns to scale.

#### Property 3: MRS and Elasticity of Substitution ( $\sigma$ )

- **Marginal Rate of Substitution:**  $R = \frac{Q_L}{Q_K} = \frac{\beta K}{\alpha L}$
- **Elasticity of Substitution:**  $\sigma = \frac{d \ln(K/L)}{d \ln(R)} = 1$  (Strictly unit elasticity for all CD functions).

## 26.2 Economic Models Using Cobb-Douglas

**Property 4: Cost Minimization** To minimize cost ( $C = rK + wL$ ) subject to a given output  $Q$ :

$$\frac{Q_L}{Q_K} = \frac{w}{r} \implies L = \frac{\beta}{\alpha} \frac{r}{w} K \quad (111)$$

Substituting this back into the production function yields the **Cost Function**, which is log-linear in output and factor prices:

$$C = kQ^{\frac{1}{n}} r^{\frac{\alpha}{n}} w^{\frac{\beta}{n}} \quad (\text{where } n = \alpha + \beta) \quad (112)$$

### Exam Point: Property 5 - Profit Maximization

**1. Perfect Competition:** Prices  $(p, r, w)$  are fixed. First Order Conditions ( $r = pQ_K$  and  $w = pQ_L$ ) yield the factor shares:

$$\alpha = \frac{rK}{pQ} \quad \text{and} \quad \beta = \frac{wL}{pQ} \quad (113)$$

*Critical Flaw:* If  $\alpha + \beta = 1$  (constant returns), the coefficient matrix is singular, making output **indeterminate**. Second Order Conditions (SOC) strictly require  $\alpha + \beta < 1$  (decreasing returns) to find a finite profit maximum.

**2. Imperfect Competition:** If the product market faces a constant elasticity demand curve  $p = bQ^\eta$ :

$$L = \frac{(\eta + 1)p\beta Q}{w} \quad \text{and} \quad K = \frac{(\eta + 1)p\alpha Q}{r} \quad (114)$$

*SOC Requirement:* The demand curve must be inelastic:  $-1 < \eta < 0$ .

## 26.3 The Statistical Model of the Production Function

**1. Cross-Section Data (Firm-Level)** To estimate parameters, a multiplicative disturbance term ( $u_{1i}$ ) is added to represent the **technical efficiency** of the firm:

$$Q_i = AK_i^\alpha L_i^\beta u_{1i} \implies \ln(Q_i) = \ln(A) + \alpha \ln(K_i) + \beta \ln(L_i) + \ln(u_{1i}) \quad (115)$$

The marginal productivity conditions also receive error terms ( $u_{2i}, u_{3i}$ ). These capture **allocative inefficiency** caused by failure to maximize profit. *Estimation Issue:* The production function suffers from an **identification problem**. Resolving this requires assuming prices are endogenous, adding partial adjustment lags, or assuming technical and economic efficiencies are uncorrelated.

**2. Time-Series Data & Technical Progress** When analyzing the same firm over time, changing relative prices and **Technical Progress** alter the function.

### Exam Point: Types of Technical Progress

**A. Disembodied Technical Progress:** Technical progress increases output independently of new investment (e.g., better management). It is modeled by adding a time trend  $e^{\theta t}$ :

$$Q_t = Ae^{\theta t} K_t^\alpha L_t^\beta \quad (116)$$

*Interpretation:* Output rises at an exponential rate of  $100 \times \theta\%$  per year.

**B. Embodied Technical Progress (Vintage Model):** Technical progress only takes effect when *new* investment occurs. A different production function applies to each "vintage" (age  $v$ ) of capital.

$$Q_t(v) = Ae^{\theta v} (K_t(v))^\alpha (L_t(v))^\beta \quad (117)$$

Total output  $Q_t$  is the integral over all past vintages:  $Q_t = \int_{-\infty}^t Q_t(v) dv$ .

## 27 Simultaneous Equations Models (SEMs)

### 27.1 The Endogeneity Problem

In standard single-equation regression, we assume regressors are independent of the error term. In a Simultaneous Equations Model (SEM), this assumption is routinely violated because variables are jointly determined.

#### Definition 15: Endogenous vs. Exogenous Variables

- **Endogenous Variables (Jointly Determined):** Variables whose values are determined within the system of equations (e.g., Price and Quantity in a supply-demand model). They act as regressors in structural equations, causing correlation with the error term.
- **Exogenous Variables (Determining):** Variables determined outside the system (e.g., Rainfall, Income). They are assumed to be strictly uncorrelated with the error terms.

*Consequence:* Ordinary Least Squares (OLS) applied directly to structural equations yields **biased and inconsistent** estimates.

## 27.2 Structural vs. Reduced Form Equations

- **Structural Equations:** The original equations formulated by economic theory (e.g.,  $q = a_1 + b_1p + c_1Y + u_1$ ).
- **Reduced-Form Equations:** Equations obtained by algebraically solving the structural equations so that each endogenous variable is expressed **only** in terms of exogenous variables and error terms (e.g.,  $q = \pi_1 + \pi_2Y + \pi_3R + v_1$ ).

*Note:* Reduced-form equations can be consistently estimated using standard OLS because all right-hand side variables are purely exogenous.

## 27.3 The Identification Problem

Identification asks: *Can we recover the true structural parameters  $(a, b, c)$  from the estimated reduced-form parameters  $(\pi)$ ?* Equivalently, can an equation be mathematically distinguished from a linear combination of all other equations in the system?

Let  $G$  be the total number of endogenous variables (which equals the total number of equations in the system).

### Theorem 32: Conditions for Identification

**1. The Order Condition (Necessary but not sufficient):** For an equation to be identified, the number of missing (excluded) variables in that equation must be at least  $(G - 1)$ .

- **Under-Identified:** Missing variables  $< (G - 1)$ . No estimates can be found.
- **Exactly Identified:** Missing variables  $= (G - 1)$ . A unique estimate exists.
- **Over-Identified:** Missing variables  $> (G - 1)$ . Multiple competing estimates exist.

**2. The Rank Condition (Necessary and Sufficient):** Delete the row of the equation being tested. Form a matrix using the columns corresponding to the variables that are **missing** from that equation. The equation is identified if and only if you can extract a non-zero determinant of size  $(G - 1) \times (G - 1)$  from this matrix. In short:

$$\text{Rank of the constructed matrix} = (G - 1) \quad (118)$$

## 27.4 Methods of Estimation

Estimation methods are broadly classified into Single Equation (Limited Information) Methods and System (Full Information) Methods.

### Exam Point: Single Equation Methods

These methods estimate one equation at a time, using restrictions from only that specific equation.

1. **Ordinary Least Squares (OLS):** Yields biased/inconsistent estimates due to endogeneity, but is computationally simple and often highly robust against specification errors.
2. **Indirect Least Squares (ILS):** Used ONLY for **exactly identified** equations.
  - *Step 1:* Estimate reduced-form equations via OLS.
  - *Step 2:* Algebraically solve for structural parameters.
3. **Two-Stage Least Squares (2SLS):** Used for **over-identified** equations (acts as a weighted average of multiple solutions).
  - *Stage 1:* Regress the problematic endogenous variable on **all** exogenous variables in the system using OLS. Obtain the predicted values ( $\hat{y}$ ).
  - *Stage 2:* Substitute the actual endogenous variable in the structural equation with its predicted value ( $\hat{y}$ ) and run OLS.

### Exam Point: System Methods & Properties

System methods estimate all equations jointly, utilizing cross-equation restrictions and the covariance of residuals (e.g., 3-Stage Least Squares (3SLS) and Full Information Maximum Likelihood (FIML)).

**Crucial Estimator Properties:**

- If all equations in a system are exactly identified, **3SLS and 2SLS estimators are mathematically identical**.
- **3SLS** is essentially a generalized, weighted Instrumental Variable (IV) estimator.
- **OLS, 2SLS, and LIML** (Limited Info Max Likelihood) are all specific members of a broader mathematical family called *k-Class estimators*.

## 27.5 Normalization in SEMs

In structural equations, we typically set the coefficient of the "dependent" endogenous variable to 1. This is called normalizing with respect to that variable (e.g.,  $y_1 - \beta y_2 - \gamma z_1 = u$ ).

- **Invariant Estimators:** Methods like FIML and LIML yield the exact same substantive results regardless of which variable you choose to normalize.
- **Variant Estimators:** 2SLS and IV methods are **not** invariant to normalization if the equation is over-identified. They only yield invariant results if the equation is exactly identified.

## 28 Vital Statistics & Demography

### 28.1 Sources of Vital Statistics

Vital Statistics deals with data relating to vital events (births, deaths, marriages, migration, etc.) in a community. The primary sources of this data are:

- **Census Data:** A comprehensive profile collected every 10 years (e.g., age, sex, occupation, literacy). *Flaw:* Fails to produce statistics for intercensal years, and birth/death enumeration is often incomplete.
- **Sample Registration System (SRS):** A large-scale demographic survey based on a **dual recording system**. It involves continuous enumeration by a resident part-time enumerator, followed by an independent six-monthly cross-verification survey.
- **Hospital Records & Ad Hoc Surveys:** Provide specialized data but fail to give a comprehensive, community-wide picture.

#### Definition 16: General Rate of a Vital Event

Rates are typically proper fractions multiplied by a constant  $K$  (usually 1,000 or 100,000) to reflect occurrences per  $K$  population.

$$\text{Rate} = \frac{\text{No. of cases of event } E \text{ in a period}}{\text{Total exposed to the risk of } E \text{ in that period}} \times K \quad (119)$$

## 28.2 Measures of Mortality

**1. Crude Death Rate (CDR):** The simplest measure of mortality. It represents the overall death rate per 1,000 people but does not account for the age or sex distribution of the population.

$$CDR = \frac{D}{P} \times 1000 \quad (120)$$

Where  $D$  = Total deaths,  $P$  = Total mid-period population.

**2. Age-Specific Death Rate (ASDR):** A death rate computed for a specific age segment, making it suitable for comparing actual mortality conditions across different communities.

$${}_n m_x = \frac{{}_n D_x}{{}_n P_x} \times 1000 \quad (121)$$

Where  ${}_n D_x$  = deaths between ages  $x$  to  $x + n - 1$  last birthday (l.b.d.), and  ${}_n P_x$  = population in that same age group. The Annual ASDR (for a 1-year interval) is simply  $m_x = \frac{D_x}{P_x} \times 1000$ .

*Relationship to CDR:* The CDR is the weighted average of ASDRs, where the weights are the population in each age group:

$$CDR = m = \frac{\sum {}_n m_x \cdot {}_n P_x}{\sum {}_n P_x} \quad (122)$$

### Exam Point: Specialized Mortality Rates

- **Infant Mortality Rate (IMR):** Deaths of children under 1 year of age.

$$IMR = \frac{D_0}{B} \times 1000$$

( $D_0$  = infant deaths,  $B$  = total live births)

- **Neonatal Mortality Rate (NMR):** Deaths of infants within the first 4 weeks (0th month) of life. Constitutes a major segment of IMR.

$$NMR = \frac{D_N}{B} \times 1000$$

- **Maternal Mortality Rate (MMR):** Deaths of mothers from puerperal causes (within 6 weeks of childbirth).

$$MMR = \frac{D_P}{B} \times 1000$$

### 28.3 The Life Table

A life table summarizes the mortality experience of a population and provides concise measures of longevity.

#### Types of Life Tables:

- **Cohort (Generation) Life Table:** Follows a specific cohort of people all born at the exact same time until the last person dies.
- **Current (Instantaneous) Life Table:** Applies the currently observed mortality rates of a given calendar period to a hypothetical cohort to model current life expectancy.

### Theorem 33: Life Table Symbols & Mathematical Relations

For a cohort starting at exact age  $x$ :

- $l_x$ : Number of persons surviving to exact age  $x$ .
- $d_x$ : Number of persons dying between exact age  $x$  and  $x + 1$ . ( $d_x = l_x - l_{x+1}$ )
- $q_x$ : Probability that a person of exact age  $x$  dies before age  $x + 1$ . ( $q_x = \frac{d_x}{l_x}$ )
- $p_x$ : Probability that a person of exact age  $x$  survives to age  $x + 1$ . ( $p_x = 1 - q_x$ )
- $L_x$ : Total aggregate years lived by the cohort between ages  $x$  and  $x + 1$ .

$$L_x = \int_0^1 l_{x+t} dt$$

- $T_x$ : Total future years lived by the  $l_x$  persons after age  $x$ .

$$T_x = \int_0^\infty l_{x+t} dt = L_x + L_{x+1} + \cdots = \sum_{i=0}^\infty L_{x+i}$$

- $e_x^0$ : Complete expectation of life (average longevity remaining) at age  $x$ .

$$e_x^0 = \frac{T_x}{l_x}$$

## 28.4 Population Dynamics

- **Stationary Population:** A population with constant size and constant age/sex composition. Annual births exactly equal annual deaths (uniformly distributed), with zero migration.
- **Stable Population:** A population with a fixed age/sex distribution, constant mortality/fertility rates, and zero migration. It grows (or shrinks) at a constant compound interest rate. *Note: A stationary population is a special case of a stable population where the overall growth rate is exactly zero.*

## 28.5 Measures of Fertility & Population Growth

**1. Crude Birth Rate (CBR):** Tracks the reproductive activity of a population over a specified period (usually a year).

$$CBR = \frac{B}{P} \times 1000 \quad (123)$$

Where  $B$  = Total live births,  $P$  = Total mid-period population.

**2. Rate of Natural Increase (RNI):** The rate of population change excluding immigration/emigration effects.

$$RNI = CBR - CDR \quad (124)$$

## Exam Point: Demographic Policies & Statistics

### Birth Rate Policies:

- **Pro-natalist Policies:** Government incentives to *increase* the birth rate (e.g., financial support in Italy).
- **Anti-natalist Policies:** Government measures to *reduce* the birth rate (e.g., China's historic one-child policy).

### Literacy Rate in India:

- Defined as the ability to acquire reading, writing, and arithmetic skills and apply them to day-to-day life.
- According to the 2011 Census, India's literacy rate was **74.04%** (estimated to be 81% by 2023-2024).

## 29 International and Indian Statistical System

### 29.1 The International Statistical System

The modern international statistical system ensures comparable measurement of economic development across countries, coordinated largely by the United Nations.

**United Nations Department of Economic and Social Affairs (UNDESA):** The main body within the UN Secretariat responsible for promoting sustainable development. Its three core functions are:

1. Compiling and analyzing economic, social, and environmental data.
2. Facilitating negotiations among Member States.
3. Advising governments and building national statistical capacities.

### Exam Point: The 10 Fundamental Principles of Official Statistics

Approved by the UN in 1994 to guide national statistical institutes:

1. **Impartiality:** Statistics must be compiled and made available on an impartial basis to honor citizens' entitlement to public information.
2. **Professional Standards:** Strict adherence to ethics and scientific methods for data collection and processing.
3. **Transparency:** Sources and methods must be presented clearly.
4. **Prevention of Misuse:** Agencies must comment on erroneous interpretations.
5. **Sources of Statistics:** Data can be drawn from surveys or administrative records, balancing quality, timeliness, and cost.
6. **Confidentiality:** Individual data must be strictly confidential and used exclusively for statistical purposes.
7. **Legislation:** Laws and regulations under which systems operate must be public.
8. **National Coordination:** Essential to achieve consistency and efficiency within a country.
9. **International Standards:** Use of international concepts and classifications promotes global consistency.
10. **International Cooperation:** Bilateral and multilateral cooperation improves global systems.

#### Global Development Goals:

- **MDGs (Millennium Development Goals):** 8 goals adopted in 2000 to address extreme poverty, hunger, and disease by 2015.
- **SDGs (Sustainable Development Goals):** 17 interlinked goals established in 2015 to continue the MDGs' work, covering broader environmental, economic, and social targets through 2030.

## 29.2 The Indian Statistical System & Administrative Data

**Administrative Data:** Data collected as a by-product of routine government administrative functions (e.g., tax records, welfare distribution).

- **Central Administrative Statistical System:** Collects statistics for sectors that straddle across multiple states (e.g., Railways, Postal Services, Telecom) or have meaning only at the all-India level (e.g., International Trade, Balance of Payments).
- **State Statistical System:** Collects all other administrative statistics strictly at the state level.

### Definition 17: Statistical Business Register (SBR)

A fundamental framework for economic statistics that provides a comprehensive list of all enterprises across India (active or closed) and their locations.

- **Purpose:** Enhances the accuracy of economic surveys, supports sectoral analysis, and aids in GDP calculations.
- **Managed By:** The Ministry of Statistics and Programme Implementation (MoSPI) and the Central Statistical Organisation (CSO).

## 29.3 Sectoral and Economic Statistics

### Sectoral Divisions:

- **Primary Sector:** Agriculture, mining, forestry, and fishing.
- **Secondary Sector:** Manufacturing industries and construction. *Note: The Annual Survey of Industries (ASI) is the key data source here.*
- **Tertiary Sector:** Services like IT, healthcare, banking, and trade.

### Key Economic Statistics:

- **Sector Accounts:** Data on income, consumption, savings, and investment for institutional sectors (households, corporations, government).
- **Price Statistics:** Consumer Price Index (CPI) and Wholesale Price Index (WPI). Crucial for understanding inflation, assessing economic performance, and guiding monetary policy.
- **Labour Market Statistics:** Data on employment, unemployment, and labour costs.

## 29.4 System of National Accounts (SNA) & ISIC

**System of National Accounts (SNA):** The internationally shared conceptual framework for measuring economic activity, published jointly by the UN, OECD, IMF, World Bank, and Eurostat (1993, revised 2008).

### Definition 18: International Standard Industrial Classification (ISIC)

The UN's hierarchical industry classification system. It categorizes production units based on the **characteristics of their economic activity** (process, technology, and output goods/services) to enable international comparison.

- **Scope:** Restricted strictly to units engaged in "economic production" as defined by the SNA.
- **Structure:** All categories at each level are mutually exclusive. It has 21 broad Sections and 88 detailed Divisions.

### Exam Point: High-Level SNA/ISIC Aggregation (A\*10 Level)

For summary data reporting in the SNA, the 21 ISIC sections are mathematically aggregated into 10 high-level categories:

| Category | ISIC Rev.4 Sections | Description                                                           |
|----------|---------------------|-----------------------------------------------------------------------|
| 1        | A                   | Agriculture, forestry and fishing                                     |
| 2        | B, C, D, E          | Manufacturing, mining/quarrying, and other industrial activities      |
| 2a       | C                   | <i>Of which: Manufacturing</i>                                        |
| 3        | F                   | Construction                                                          |
| 4        | G, H, I             | Wholesale/retail trade, transportation/storage, accommodation/food    |
| 5        | J                   | Information and communication                                         |
| 6        | K                   | Financial and insurance activities                                    |
| 7        | L                   | Real estate activities                                                |
| 8        | M, N                | Professional, scientific, technical, administrative, support services |
| 9        | O, P, Q             | Public administration/defence, education, human health/social work    |
| 10       | R, S, T, U          | Other service activities                                              |

## Practical Exercise 7: Solution

**Problem Statement:** Apply the graphical test of lognormality to examine whether the underlying true distributions of PCE are approximately two-parameter lognormal.

### Solution Step 1: Theoretical Foundation

To test if empirical grouped data follows a two-parameter Lognormal distribution  $LN(\mu, \sigma^2)$ , we utilize the standard normal deviate (normit).

For a lognormal distribution, the cumulative probability  $P$  up to income  $x$  is given by:

$$P = \Phi\left(\frac{\ln x - \mu}{\sigma}\right) \quad (125)$$

where  $\Phi$  is the standard normal cumulative distribution function (CDF). Let  $t_P$  be the standard normal deviate such that  $\Phi(t_P) = P$ . We can express this as a linear equation:

$$t_P = -\frac{\mu}{\sigma} + \frac{1}{\sigma} \ln(x) \quad (126)$$

**The Test:** If we plot the natural logarithm of the upper class boundaries ( $\ln x$ ) on the X-axis against the corresponding standard normal deviates ( $t_P$ ) on the Y-axis, the points should fall approximately on a straight line with slope  $1/\sigma$  and intercept  $-\mu/\sigma$ .

### Solution Step 2: The Calculation Table

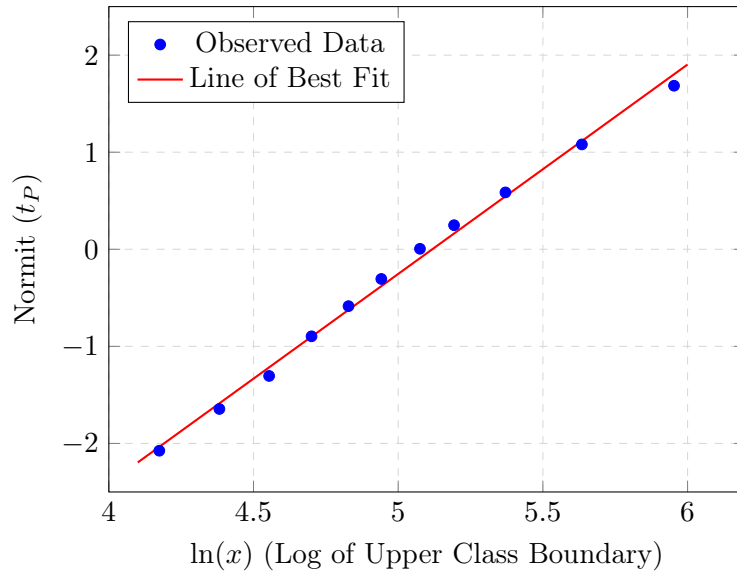
We calculate the cumulative frequencies ( $cf$ ) up to the upper boundary of each class (excluding the open-ended final class), find the cumulative relative frequencies ( $P$ ), and use a standard normal table to find the normits ( $t_P$ ).

### Solution Step 3: The Graphical Plot

Plotting the data points ( $\ln x, t_P$ ) on a standard Cartesian coordinate system.

| PCE Class<br>(Rs.) | Upper Bound<br>( $x$ ) | Log Upper Bound<br>( $\ln x$ ) | Frequency<br>( $f_i$ ) | Cum. Freq.<br>( $cf_i$ ) | Cum. Prob.<br>( $P = cf_i/1000$ ) | Normit<br>( $t_P$ ) |
|--------------------|------------------------|--------------------------------|------------------------|--------------------------|-----------------------------------|---------------------|
| 000-065            | 65                     | 4.1744                         | 19                     | 19                       | 0.019                             | -2.0749             |
| 065-080            | 80                     | 4.3820                         | 31                     | 50                       | 0.050                             | -1.6449             |
| 080-095            | 95                     | 4.5539                         | 46                     | 96                       | 0.096                             | -1.3047             |
| 095-110            | 110                    | 4.7005                         | 89                     | 185                      | 0.185                             | -0.8965             |
| 110-125            | 125                    | 4.8283                         | 94                     | 279                      | 0.279                             | -0.5858             |
| 125-140            | 140                    | 4.9416                         | 101                    | 380                      | 0.380                             | -0.3055             |
| 140-160            | 160                    | 5.0752                         | 122                    | 502                      | 0.502                             | 0.0050              |
| 160-180            | 180                    | 5.1930                         | 96                     | 598                      | 0.598                             | 0.2482              |
| 180-215            | 215                    | 5.3706                         | 123                    | 721                      | 0.721                             | 0.5858              |
| 215-280            | 280                    | 5.6348                         | 139                    | 860                      | 0.860                             | 1.0803              |
| 280-385            | 385                    | 5.9532                         | 94                     | 954                      | 0.954                             | 1.6849              |
| 385- $\infty$      | —                      | —                              | 46                     | <b>1000</b>              | —                                 | —                   |

Graphical Test of Lognormality (Rural PCE)



## Conclusion

As clearly visible in the graph above, the plotted coordinates of  $(\ln x, t_P)$  fall almost perfectly along a straight line (the calculated coefficient of determination  $R^2 \approx 0.996$ ).

Because the empirical data strictly conforms to the linear theoretical relationship  $t_P = -\frac{\mu}{\sigma} + \frac{1}{\sigma} \ln(x)$ , we can conclusively state that the underlying true distribution of Per Capita Expenditure (PCE) for Rural India is **approximately a two-parameter lognormal distribution**.

## Practical Exercise 8: Solution

**Problem Statement:** Apply the method of quantiles to estimate the parameters of the underlying lognormal distributions for Rural India PCE. Hence, compute the Lorenz Ratio.

### Solution Step 1: Methodology

For a two-parameter lognormal distribution  $LN(\mu, \sigma^2)$ , the income  $x$  corresponding to a cumulative probability  $P$  is given by:

$$\ln(x_P) = \mu + \sigma t_P \quad (127)$$

where  $t_P$  is the standard normal deviate (normit) such that  $\Phi(t_P) = P$ .

To estimate  $\mu$  and  $\sigma$  using the **Method of Quantiles**, we select two points  $(x_1, P_1)$  and  $(x_2, P_2)$  from the cumulative distribution. To maximize efficiency, symmetric quantiles (e.g., the 27th and 73rd or 30th and 70th percentiles) are preferred[cite: 4851, 4852].

The parameters are then calculated as:

$$\hat{\sigma} = \frac{\ln(x_2) - \ln(x_1)}{t_2 - t_1} \quad (128)$$

$$\hat{\mu} = \ln(x_1) - \hat{\sigma} t_1 \quad (129)$$

### Solution Step 2: The Calculation Table

We first convert the frequency distribution into a cumulative distribution to identify quantile points.

| PCE Class Upper Bound ( $x$ ) | Frequency ( $f$ ) | Cum. Freq. ( $cf$ ) | Cum. Prob. ( $P$ ) | Normit ( $t_P$ ) |
|-------------------------------|-------------------|---------------------|--------------------|------------------|
| 65                            | 19                | 19                  | 0.019              | -2.0749          |
| 80                            | 31                | 50                  | 0.050              | -1.6449          |
| 95                            | 46                | 96                  | 0.096              | -1.3047          |
| 110                           | 89                | 185                 | 0.185              | -0.8965          |
| 125                           | 94                | 279                 | 0.279              | -0.5858          |
| <b>140</b>                    | <b>101</b>        | <b>380</b>          | <b>0.380</b>       | <b>-0.3055</b>   |
| 160                           | 122               | 502                 | 0.502              | 0.0050           |
| 180                           | 96                | 598                 | 0.598              | 0.2482           |
| <b>215</b>                    | <b>123</b>        | <b>721</b>          | <b>0.721</b>       | <b>0.5858</b>    |
| 280                           | 139               | 860                 | 0.860              | 1.0803           |
| 385                           | 94                | 954                 | 0.954              | 1.6849           |

### Solution Step 3: Parameter Estimation

We select the following two symmetric quantiles from the table:

- **Quantile 1:**  $x_1 = 140$ ,  $P_1 = 0.380$ ,  $t_1 = -0.3055$

- **Quantile 2:**  $x_2 = 215$ ,  $P_2 = 0.721$ ,  $t_2 = 0.5858$

**1. Estimating  $\sigma$ :**

$$\begin{aligned}\hat{\sigma} &= \frac{\ln(215) - \ln(140)}{0.5858 - (-0.3055)} \\ \hat{\sigma} &= \frac{5.3706 - 4.9416}{0.8913} = \frac{0.4290}{0.8913} \\ \hat{\sigma} &\approx \mathbf{0.4813}\end{aligned}$$

**2. Estimating  $\mu$ :**

$$\begin{aligned}\hat{\mu} &= \ln(140) - (0.4813 \times -0.3055) \\ \hat{\mu} &= 4.9416 + 0.1471 = \mathbf{5.0887}\end{aligned}$$

**Solution Step 4: Computing Lorenz Ratio**

For a lognormal distribution, the Lorenz Ratio (Gini Coefficient) is determined entirely by the parameter  $\sigma$ [cite: 5046, 5047]:

$$LR = 2\Phi\left(\frac{\sigma}{\sqrt{2}}\right) - 1 \quad (130)$$

Substituting our estimated  $\hat{\sigma} = 0.4813$ :

$$\begin{aligned}LR &= 2\Phi\left(\frac{0.4813}{1.4142}\right) - 1 \\ LR &= 2\Phi(0.3403) - 1\end{aligned}$$

From standard normal tables,  $\Phi(0.34) \approx 0.6332$ .

$$\begin{aligned}LR &= 2(0.6332) - 1 \\ LR &= 1.2664 - 1 = \mathbf{0.2664}\end{aligned}$$

**Conclusion:** The estimated parameters for the underlying lognormal distribution of Rural India PCE are  $\hat{\mu} = 5.0887$  and  $\hat{\sigma} = 0.4813$ . The corresponding Lorenz Ratio is **0.2664**.

**Practical Exercise 9: Solution**

**Problem Statement:** Compute (i) Head Count Ratio (H), (ii) Income Gap Ratio (I), and (iii) Sen Index of Poverty ( $P_s$ ) from the given Rural Sector PCE data for 1983.

**Solution Step 1: Determining the Poverty Line (Z)**

The poverty line is given as Rs. 15 per 30 days at 1960-61 prices[cite: 6791]. To find the poverty line for the current period (1983), we adjust it using the Consumer Price Index (CPI)[cite: 6791]:

$$\begin{aligned}Z &= \text{Poverty Line}_{1960-61} \times \frac{\text{CPI}_{1983}}{\text{CPI}_{1960-61}} \\ Z &= 15 \times \frac{511}{100} = \mathbf{76.65}\end{aligned}$$

Any individual with an income (PCE) below Rs. 76.65 is considered poor[cite: 5117, 5132].

**Solution Step 2: Identification of the Poor**

Based on the calculated  $Z = 76.65$ , we identify the population groups in Table 1 that fall below this threshold. The first five classes (00 – 30 to 60 – 70) have average PCEs strictly below  $Z$ .

| PCE Class (Rs.)   | Pop. Percentage ( $f_i$ )   | Average PCE ( $y_i$ )        |
|-------------------|-----------------------------|------------------------------|
| 00 – 30           | 0.91                        | 24.80                        |
| 30 – 40           | 2.48                        | 35.79                        |
| 40 – 50           | 5.10                        | 45.42                        |
| 50 – 60           | 7.98                        | 55.23                        |
| 60 – 70           | 9.75                        | 65.15                        |
| <b>Total Poor</b> | $\sum f_i = \mathbf{26.22}$ | $\bar{y}_q = \mathbf{54.12}$ |

### Solution Step 3: Calculating Poverty Measures

**(i) Head Count Ratio (H):** The proportion of the population below the poverty line[cite: 5141, 5142].

$$H = \frac{q}{n} = \frac{26.22}{100} = \mathbf{0.2622} \quad (131)$$

**(ii) Income Gap Ratio (I):** The average poverty gap of the poor as a fraction of the poverty line[cite: 5149, 5153]. Let  $\bar{y}_q$  be the average income of the poor (Rs. 54.12)[cite: 5151, 5152].

$$I = 1 - \frac{\bar{y}_q}{Z}$$

$$I = 1 - \frac{54.12}{76.65} = 1 - 0.7060 = \mathbf{0.2940}$$

**(iii) Sen Index of Poverty ( $P_s$ ):** Incorporates  $H$ ,  $I$ , and the inequality among the poor ( $G$ )[cite: 5189, 5190]. For this dataset, the Gini coefficient of the poor ( $G$ ) is approximately 0.1093.

$$P_s = H[I + (1 - I)G]$$

$$P_s = 0.2622[0.2940 + (1 - 0.2940) \times 0.1093]$$

$$P_s = 0.2622[0.2940 + 0.0772] = 0.2622 \times 0.3712 = \mathbf{0.0973}$$

**Conclusion:** For Rural India in 1983, the Head Count Ratio was **26.22%**, the average depth of poverty (Income Gap Ratio) was **0.2940**, and the overall degree of poverty (Sen Index) was **0.0973**.

## Practical Exercise 10: Solution

**Problem Statement:** Calculate the (i) Coefficient of Variation (CV), (ii) Relative Mean Deviation (RMD), and (iii) Lorenz Ratio of incomes of  $N$  individuals with incomes  $1, 2, 3, \dots, N$  units, respectively. What will happen if  $N$  goes to infinity?

### Solution Step 1: Preliminaries (Means and Sums)

Given the sequence of incomes  $\{1, 2, 3, \dots, N\}$ , we first calculate the necessary statistical sums.

- **Sum of Incomes:**  $\sum_{i=1}^N i = \frac{N(N+1)}{2}$
- **Arithmetic Mean ( $\bar{y}$ ):**  $\bar{y} = \frac{1}{N} \sum i = \frac{N+1}{2}$  [cite: 2398]
- **Sum of Squares:**  $\sum_{i=1}^N i^2 = \frac{N(N+1)(2N+1)}{6}$

### Solution Step 2: (i) Coefficient of Variation (CV)

The CV is defined as the ratio of the Standard Deviation (SD) to the Arithmetic Mean (AM)[cite: 2400].

#### 1. Variance ( $\sigma^2$ ):

$$\begin{aligned}\sigma^2 &= \frac{1}{N} \sum i^2 - \bar{y}^2 \\ &= \frac{(N+1)(2N+1)}{6} - \left(\frac{N+1}{2}\right)^2 \\ &= \frac{N+1}{2} \left[ \frac{2N+1}{3} - \frac{N+1}{2} \right] \\ &= \frac{N+1}{2} \left[ \frac{4N+2-3N-3}{6} \right] = \frac{N^2-1}{12}\end{aligned}$$

#### 2. CV calculation:

$$CV = \frac{\sqrt{\frac{N^2-1}{12}}}{\frac{N+1}{2}} = \sqrt{\frac{(N-1)(N+1)}{12} \cdot \frac{4}{(N+1)^2}} = \sqrt{\frac{N-1}{3(N+1)}} \quad (132)$$

As  $N \rightarrow \infty$ :  $CV \rightarrow \sqrt{\frac{1}{3}} \approx 0.5774$

### Solution Step 3: (ii) Relative Mean Deviation (RMD)

The RMD is defined as  $\frac{1}{n\bar{y}} \sum |y_i - \bar{y}|$ [cite: 2397]. Since  $\bar{y} = \frac{N+1}{2}$ , for even  $N = 2k$ , the mean falls between  $k$  and  $k+1$ . For simplicity, we look at the limit. The sum of absolute deviations  $\sum |i - \frac{N+1}{2}|$  behaves approximately like  $\int_0^N |x - \frac{N}{2}| dx = \frac{N^2}{4}$ .

**Calculation for Large  $N$ :**

$$RMD \approx \frac{N^2/4}{N \cdot (N/2)} = \frac{N^2/4}{N^2/2} = 0.50$$

*Detailed discrete result:* For large  $N$ , the RMD approaches **0.5**.

### Solution Step 4: (iii) Lorenz Ratio (LR)

We use the simplified formula for sorted discrete data:  $LR = 1 - \frac{1}{n^2\bar{y}} \sum_{i=1}^n \{2(n-i)+1\}y_i$ [cite: 2405, 2460]. Let  $y_i = i$  (already sorted  $1 \leq 2 \leq \dots \leq N$ ).

$$\begin{aligned}\sum_{i=1}^N \{2(N-i)+1\}i &= (2N+1) \sum i - 2 \sum i^2 \\ &= (2N+1) \frac{N(N+1)}{2} - 2 \frac{N(N+1)(2N+1)}{6} \\ &= (2N+1) \frac{N(N+1)}{2} \left[ 1 - \frac{2}{3} \right] \\ &= \frac{N(N+1)(2N+1)}{6}\end{aligned}$$

**Substitute into LR formula:**

$$\begin{aligned}LR &= 1 - \frac{\frac{N(N+1)(2N+1)}{6}}{N^2 \frac{N+1}{2}} \\ &= 1 - \frac{2N+1}{3N} = \frac{3N-2N-1}{3N} = \frac{N-1}{3N}\end{aligned}$$

As  $N \rightarrow \infty$ :  $LR \rightarrow \frac{1}{3} \approx 0.3333$

## Summary of Results

| Measure                       | Formula for Finite $N$      | Limit as $N \rightarrow \infty$ |
|-------------------------------|-----------------------------|---------------------------------|
| Coefficient of Variation (CV) | $\sqrt{\frac{N-1}{3(N+1)}}$ | $1/\sqrt{3} \approx 0.5774$     |
| Relative Mean Deviation (RMD) | $\approx 0.5$               | <b>0.5000</b>                   |
| Lorenz Ratio (LR)             | $\frac{N-1}{3N}$            | $1/3 \approx 0.3333$            |

## 30 End Semestral Examination: Questions and Solutions

### Part A: Quantitative Problems

**Question 1:** Suppose monthly per capita expenditures ( $X$ ) follow Pareto law with inequality parameter  $\alpha = 2$  and threshold parameter  $c = \text{Rs. } 100$ . Find:

1. The average monthly per capita expenditure of all individuals.
2. The average monthly PCE of individuals spending less than Rs. 200.
3. The average monthly PCE of individuals spending more than Rs. 200.

### Solution to Question 1

The Pareto PDF is  $f(x) = \alpha c^\alpha x^{-\alpha-1}$  for  $x \geq c$  [cite: 9]. Given  $\alpha = 2$  and  $c = 100$ .

1. **Average PCE of all individuals:** Using the mean formula  $\mu = \frac{\alpha c}{\alpha-1}$  [cite: 9]:

$$\mu = \frac{2 \times 100}{2-1} = \frac{200}{1} = \text{Rs. } 200 \quad (133)$$

2. **Average PCE for  $X < 200$ :** We calculate the conditional expectation  $E(X|X < 200) = \frac{1}{F(200)} \int_{100}^{200} x f(x) dx$ .

- $F(200) = 1 - (100/200)^2 = 1 - 0.25 = 0.75$  [cite: 9].
- $\int_{100}^{200} x(2 \cdot 100^2 \cdot x^{-3}) dx = 20000 \int_{100}^{200} x^{-2} dx = 20000[-x^{-1}]_{100}^{200} = 20000(0.01 - 0.005) = 100$ .
- $E(X|X < 200) = 100/0.75 = \text{Rs. } 133.33$

3. **Average PCE for  $X > 200$ :** Due to the properties of the Pareto distribution, the tail above any point  $x_0$  is also a Pareto distribution with threshold  $x_0$  [cite: 9]:

$$E(X|X > 200) = \frac{\alpha \cdot 200}{\alpha-1} = \frac{2 \times 200}{1} = \text{Rs. } 400 \quad (134)$$

**Question 2:** Suppose in a population the mean income, Lorenz Ratio, and number of people below Rs. 5,000 are Rs. 3500, 0.2, and 100 respectively. For the group above Rs. 5,000, these values are Rs. 12500, 0.4, and 500 respectively. Find the overall Lorenz ratio.

## Solution to Question 2

We use the Gini/Lorenz Ratio decomposition for non-overlapping groups. Let  $n_1 = 100, \mu_1 = 3500, L_1 = 0.2$  and  $n_2 = 500, \mu_2 = 12500, L_2 = 0.4$ .

### Step 1: Calculate Shares

- Total population  $n = n_1 + n_2 = 600$ . Population shares:  $w_1 = 1/6, w_2 = 5/6$ .
- Total income  $Y = (100 \times 3500) + (500 \times 12500) = 350,000 + 6,250,000 = 6,600,000$ .
- Income shares:  $s_1 = 350,000/6,600,000 = 35/660 \approx 0.053$ ;  $s_2 = 6,250,000/6,600,000 = 625/660 \approx 0.947$ .

**Step 2: Apply Decomposition Formula**  $L_{overall} = w_1 s_1 L_1 + w_2 s_2 L_2 + (s_2 - w_2)$ [cite: 10]:

$$L = (0.1667 \times 0.053 \times 0.2) + (0.8333 \times 0.947 \times 0.4) + (0.947 - 0.8333)$$

$$L = 0.00177 + 0.31565 + 0.1137 = \mathbf{0.4311}$$

**Question 3:** Compute (i) Head Count Ratio (H), (ii) Income Gap Ratio (I), and (iii) Sen Index of Poverty (P) from the data provided. Assume the poverty line  $Z = \text{Rs. } 630$ .

## Solution to Question 3

**Step 1: Identification of Poor** Individuals are poor if their Average PCE  $< Z = 630$ . This applies to the first five classes (000 – 200 to 500 – 600).

### Step 2: Calculations

1. **Head Count Ratio (H):** Sum of percentages for poor classes[cite: 11].

$$H = (0.87 + 2.25 + 4.74 + 6.99 + 8.50)/100 = \mathbf{0.2335} \quad (135)$$

2. **Income Gap Ratio (I):**  $1 - (\bar{y}_q/Z)$ [cite: 11]. Weighted mean income of poor  $\bar{y}_q = \frac{\sum(\% \times \text{Avg})}{\sum \%} = \frac{10525.65}{23.35} = 450.78$ .

$$I = 1 - (450.78/630) = \mathbf{0.2844} \quad (136)$$

3. **Sen Index (P):**  $P = H[I + (1 - I)G]$ [cite: 11]. Using the calculated Gini of poor  $G \approx 0.1290$ :

$$P = 0.2335[0.2844 + (0.7156 \times 0.1290)] = \mathbf{0.0880} \quad (137)$$

## Part B: Theoretical Concepts

**Question 4:** Write down the desirable properties of a measure of inequality. Give examples of positive measures. Outline Atkinson's Social Welfare Function approach.

### Answer to Question 4

**Desirable Properties:** Scale Independence, Pigou-Dalton Principle of Transfers, and the Principle of Diminishing Transfers[cite: 9, 10]. **Positive Measures:** Relative Mean Deviation, Coefficient of Variation, and Lorenz Ratio[cite: 9]. **Atkinson's Approach:** Atkinson introduced the **Equally Distributed Equivalent Income** ( $y_{ede}$ ), which is the income level that, if given to everyone equally, would yield the same total social welfare as the current distribution[cite: 10]. His index is  $I_A = 1 - (y_{ede}/\bar{x})$ [cite: 10].

**Question 5:** Define Head Count Ratio and Income Gap Ratio. Discuss merits/demerits and Sen's improvement.

### Answer to Question 5

- **H (Head Count Ratio):** Proportion of population below  $Z$ . *Merit:* Simple. *Demerit:* Ignores depth of poverty[cite: 11].
- **I (Income Gap Ratio):** Average depth of poverty. *Merit:* Measures depth. *Demerit:* Ignores how many are poor[cite: 11].
- **Sen's Improvement:** Sen introduced the **Weak Transfer Axiom**, requiring the index to increase if income is transferred from a poorer person to a richer person below  $Z$ . His index  $P_s$  combines  $H$ ,  $I$ , and  $G$  (inequality among poor) to satisfy this[cite: 11].

**Question 6:** Define Cobb-Douglas (CD) production function. State properties. Describe cost minimization and profit maximization.

### Answer to Question 6

**Definition:**  $Q = AK^\alpha L^\beta$ [cite: 16]. **Properties:** Homogeneous of degree  $(\alpha + \beta)$ ; positive but diminishing marginal products ( $0 < \alpha, \beta < 1$ ); unit elasticity of substitution ( $\sigma = 1$ )[cite: 16]. **Optimization:**

- **Cost Minimization:** Set MRS equal to the factor price ratio:  $f_L/f_K = w/r$ [cite: 15, 16].
- **Profit Maximization:** Set value of marginal products equal to factor prices:  $pf_K = r$  and  $pf_L = w$ [cite: 15, 16].

**Question 7:** Describe Prais-Houthakker (PH) formulation for household scale economies.

### Answer to Question 7

The PH model assumes a household of size  $n$  has an **effective size**  $n^{\theta_i}$  for specific commodity  $i$  and  $n^{\theta_0}$  for total income[cite: 14]. The equation is  $\frac{y_i}{n^{\theta_i}} = f\left(\frac{x}{n^{\theta_0}}\right)$ [cite: 14]. **Economy of Scale** exists if  $\theta < 1$ , occurring when consumption (e.g., electricity or bulk food) rises less than proportionately with household size[cite: 14].